ROYAL CANADIAN NAVY QUALIFICATION STANDARD AND PLAN (QSP)

Naval Warfare Officer (NWO)
(00207.1.N.05, 00207.1.N.06))

Advanced Arctic Navigation (AAN)
ALIF

MITE Course IDs:
Residential AAN (ALIF): 122173

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RECORD OF AMENDMENTS

The "Authority" reference indicates approval of this QSP document. Authority for chapter 1 of this QSP resides with the Commander Naval Personnel and Training Group (CNPTG). The authority for chapters 2 and 3 of this QSP Commanding Officer Naval Training Development Centre (Pacific) (CO NTDC (P)), who is both the Training Authority (TA) and the Curriculum Control Authority (CCA). The approval date appears in the bottom right corner of every page (the change date shall be updated in the footer). The Revision column is intended to provide a short indication of where the change(s) was/were made. The history of amendments to previous QSPs is detailed in Annex J "Course History" and shall henceforth be updated with any changes to this version (V1.5) of the QSP. See Annex C for Record of Decisions (ROD).

Table 1 – Record of Amendments

REVISION	APPROVAL DATE	AUTHORITY
1.0 QSPWB Feb 2018		CNPTG
1.5 Revisions Jul 2020		CNPTG
		NTDC (P) CO

FOREWORD

1. This training document is developed under the authority of the Chief Maritime Staff Commander Naval Personnel and Training Group (CNPTG) and in accordance with the following references:

   a. A-P9-050 Series, Canadian Forces Manual of Individual Training and Education System (CFITES);

   b. A-PD-055-002/PP-003, Canadian Armed Forces Military Employment Structure (MES), Volume 2 – General Specifications, Part 3 – Officer General Specifications dated 23 Apr 2014;

   c. A-PD-055-002/PP-001, Name of Occupation Job Based Specification or Job Based Occupational Specification dated DD Month YY;

   d. A-PD-055-003/PQ-001 Occupation Specialty Specification – Fleet Navigating Officer (FNO)

   e. A-PD-055-003/PQ-001 Occupation Specialty Specification – Advanced Arctic and Offshore Patrol Vessel Operator (OSS ALIF – dated 21 Feb 2017);

   f. NPTORD 4500-22 Naval Personnel Training Order – Qualification Standard & Plan.

2. This qualification standard and plan (QSP) publication is effective on approval and receipt superseding all previous versions including FNO Qualification (to Advanced Arctic Navigation and is meant to follow the new version of ALIB (to be updated as Basic Arctic Navigation (BAN).

3. Performance Objectives (POs) and Enabling Objectives (EOs): POs and EOs which specify the standards that students must achieve to become qualified are listed in the Table of Contents and are explained fully in Chapter 3.

4. Assessment Details: Assessment details for all Enabling Checks (ECs), Critical Enabling Checks (CECs) and Performance Checks (PCs) can be found in Annex K.

5. Amendments: Suggestions for amendments may be initiated by anyone and must be forwarded through the normal chain of command summarizing all requested amendments in writing to the applicable approving authority. IAW NPTORD 4500-22 Table 6.1, minor amendments to Chapter 2 and 3 are to be forwarded to course control document managers (CCDM) at NTDC (P) and major amendments and final approval for Chapter 1 amendments are to be submitted to the CNPTG.

6. Description of Need: To ensure the professional navigation of RCN vessels, a requirement exists for a specialist officer to carry out the duties of a Navigation Officer for Arctic and Offshore Patrol Vessels (AOPV) through Ice Covered Waters.

7. Method of Achieving Objectives: The objectives of the course are met by a combination of theoretical and experiential training and performance-based assessments in the classroom and in simulators.

8. This QSP shall be used by NFS(P) as the primary authority governing development, conduct, program evaluation and validation of the Advanced Arctic Navigation course.

9. This QSP must be used in conjunction with Occupation Specialty Specifications ALIR and ALIF in A-PD-055-003/PQ-001.

10. The designated TEs for this training are indicated below:

Table 2 - Training Establishments

TE	Phase / Module	MITE Course ID	NQual
NFS(P)	All	122173	ALIF

CHAPTER 1 - TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY
COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (CNPTG)

AIM

1. To ensure that a professional manner of maritime navigation is conducted throughout the Canadian Armed Forces, a requirement exists for a specialist officer to execute the navigation of Royal Canadian Naval Vessels in Ice Covered Waters. The Qualification Standard and Plan (QSP) establishes the requirements and training for the Advanced Arctic Navigation (AAN) Course (ALIF/122173). This QSP is designed to assist NTDC (P) in developing, NFS (A or P) in conducting, and NPTG in evaluating and validating the training. This course is designed to expand upon the knowledge and experience previously acquired by Naval Warfare Officers (NWOs).

COURSE DURATION

2. Course duration is calculated in accordance with NPTORD 4500-22 Annex B. Detailed sequencing and timings are found at Annex G of this QSP. Table 3 below, details the overall estimated allocation of timing for this course. The estimation is based on the NPTORD 4500-22 direction for allocation of physical training and administrative time.

Table 3 - Course Duration

Allocation of ATT	**Estimated Time in Minutes**
Enabling Objectives / Teaching Points	1390
Enabling Checks (ECs)	210
Critical Enabling Checks (CECs)	300
ATT Sub-total	**1900**
Allocation of BAT, TAT, TTT, PT (Course > 31 days)	
Base Administrative Time (BAT)	50
Training Administrative Time (TAT)	50
Training Travel Time (TTT)	0
Physical *Fitness* Training (PT)	0
Sub-total	**100**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT)	2000
Total mins/400 = # days (rounded up)	5
Total Training Days	**5**

RESOURCES

3. Required Course Staffing: Table 4 below, details key personnel required to conduct the training:

Table 4 - Required Course Staffing

Required Course Staffing					
Position	Minimum Rank	Qualifications Required	Qty	Remarks	PC/EC, PO/EO
Mentor	LCdr	NWO (MOSID 00207), ALIF AEEA, AEEG preferred	1	LCdr or Cdr rank Surface Command Part II qualified, Ice experience in 4/10th ice or greater for 50 days	All
Instructor	Lt (N)	NWO (MOSID 00207), ALIF AEEA, ALPQ	1	Post director tour FNO	All
Lecturer	CCG	CCG Icebreaker	1	Current serving Commanding Officer or Chief Officer of a CCG icebreaker preferred	All

4. Critical Training Resources: Table 5 below, details the critical training resources to conduct the training. It should be read in conjunction with the recommended course timetable in Annex G and the provision of additional training resources required in Annex H of this QSP.

Table 5 - Critical Training Resources

Critical Training Resources	Training Days	PO/EO
FNO Classroom	Full Course	All
Smart Board	Full Course	All
DWAN Desktop connected to Smart Board	Full Course	All

Critical Training Resources	Training Days	PO/EO
Colour Printer	Full Course	All
Navigation and Bridge Simulator (NABS)	Full Course	All

COURSE CAPACITY

5. Planning for the conduct of training is based on the guidelines in CFITES Vol 5 Annex D (Instructor to Learner Ratios). Justification can be found in Annex C – Record of Proceedings (ROP).

   a. Maximum – 6;
   b. Minimum – 3; and
   c. Optimum – 3.

COURSE LOADING

6. Loading: The authority to load members onto this course lies with the NFS (P). NPTG is to be included as an INFO addressee on all loading messages.

PREREQUISITES

7. The loading authority must ensure that members have the associated prerequisites listed in table 6 , before being selected for this course:

Table 6 - Prerequisites for Loading on Course

Fleet Navigating Officer	**Executive Officer**	**Commanding Officer**
Appointment to AOPV as FNO	Appointment to AOPV as XO	Appointment to AOPV as CO
Qualified NWO IV 00207	Qualified NWO IV 00207	Qualified NWO IV 00207
Qualified ALIB (Basic Arctic Navigation (BAN))	Qualified ALIB (Basic Arctic Navigation (BAN))	Qualified ALIB (Basic Arctic Navigation (BAN))
Security Clearance Level III	Security Clearance Level III	Security Clearance Level III
Substantive Rank of Lieutenant (Navy) (Lt(N))	Substantive Rank of Lieutenant Commander (LCdr)	Substantive Rank of Commander (Cdr)

SCHEDULING

8. Scheduling: Each session of this course is to be scheduled in MITE by the NFS conducting the training and NPTG through the Quantity Control section. The course length in MITE is to be based on the course duration, planned annual leave, training breaks, weekends and training system holidays. Course sessions for Naval Reserve Force may be scheduled during training system holidays.

MITE

9. MITE Qualification: Successful completion of this course constitutes the primary means of attaining the Advanced Arctic Navigation (AAN) Course (NQual ALIF) with the associated MITE ID: 112173.

COURSE REPORTING

10. Successful trainees shall receive a Course Report (CF 377) and Certificate of Military Achievement (CF 289) completed IAW DAOD 5031-9 "Course Reporting and Certificates" and NPTORD 4500-51 "Course Reports and Certification".

11. Unsuccessful trainees shall receive a Course Report (CF377) only.

EVALUATION AND VALIDATION

12. Program evaluation shall be conducted IAW NPTG HQ Dev Sched and CFITES Vol. 11 "Evaluation of Instructional Programmes".

13. Program validation shall be conducted IAW NPTG HQ Dev Sched and CFITES Vol. 8 "Validation of Instructional Programmes".

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

14. PLARs shall be conducted IAW NAVORD 4500-2 "RCN Prior Learning Assessment and Recognition", DAOD 5031-1 "Canadian Forces Military Equivalencies Program", and CFITES Vol 12 "Canadian Forces Military Equivalencies Programme: Prior Learning Assessment".

SPECIAL INSTRUCTIONS

15. Nil.

CHAPTER 2 – ASSESSMENT

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE NAVAL TRAINING DEVELOPMENT CENTRE (PACIFIC)

GENERAL

1. Philosophy:

2. This is a pass/fail course. Students must successfully complete all POs to pass this course. Performance Checks (PCs), Critical Enabling Checks (CECs) and Enabling Checks (ECs) are based on the Performance Objectives (POs) and Enabling Objectives (EOs) in Chapter 3 of this QSP. All PCs must be successfully completed to achieve the qualification.

3. To be successful, students must consistently conduct navigation in all environmental conditions and traffic density levels, adjust course and/or speed as required, assess positional accuracy, and provide Command with accurate and timely navigation and shiphandling advice. It is crucial that navigators consistently maintain their integrity and honesty at all times with the CO/OIC.

4. Students will be developed and assessed using a combination of formative and summative assessment events. At a minimum, these will involve PCs, ECs, end of lesson confirmation, homework, assignments, and/or quizzes. See Annex K for details.

5. Dress and deportment, ethics, conduct and behaviour demonstrated by the trainee shall be evaluated and assessed based on the standards laid out under Operation Honour and the LRH program. A trainee who displays inappropriate dress and deportment, ethics, and/or behaviour is considered to have demonstrated unsatisfactory conduct which may result in the convening of a Training Review Board (TRB).

ENABLING AND PERFORMANCE CHECKS

6. Enabling Checks (ECs): ECs are formative assessments that help the trainee and the instructors recognize progress or gaps in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning to improve retention of course content. Formative assessments are designed to provide feedback to both instructional staff and trainees. Each EO must be formatively assessed prior to summative assessments.

7. In the event of an EC failure, the trainee will be counselled and given remedial training and/or additional homework, as appropriate. In the event of a consecutive EC failure, the trainee will be counselled IAW CFITES Vol 6 with the unsatisfactory progress documented in their Student File.

8. Critical Enabling Checks (CECs): CECs are summative assessments at the end of an EO which are considered critical because trainees cannot progress in the

training without first successfully completing them. They are related to elements such as safety and security where harm could result from poor performance. CECs are treated the same as PCs.

9. Performance Checks (PCs): PCs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

10. PCs will be conducted after formative assessment has occurred and the trainee has had the opportunity to develop skills and apply knowledge related to the PO.

PROGRESS MONITORING

11. Student progress will be monitored using formative assessment tools to help the trainee and instructional staff recognize progress or lapses in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. Formative assessment is used during a phase of instruction to reinforce learning, so that content is retained.
12. Monitoring learning is the responsibility of all instructional staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW TE SOs or SOPs and CFITES Vol 6 "Conduct of Instruction Programmes".
13. Continuous monitoring of student progress is required to provide the following:
    a. Early warning of trainee difficulties;
    b. Feedback on the effectiveness of training;
    c. Information for any required TRBs ; and
    d. Information to assist in completion of the CF 377 Course Report.
14. Students experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW TE SOs or SOPs. A record of trainee counselling is to be maintained by the training establishment.
15. Instructional staff shall maintain individual trainee files, including the results of all:
    a. Student performance documentation (e.g. Weekly write-up, medical chits, dress inspection chits, conduct chits, etc.);
    b. Formative assessment material (e.g. ECs, practice periods, homework, assignments and quizzes, etc.); and
    c. Summative assessment material (PCs, CECs).

TRAINING REVIEW BOARD

16. All PC failures and supplemental failures are to be handled IAW NPTORD 4500-54 "Training Review Board Process". Supplemental PCs will normally be scheduled not less than 24 hours post counselling, depending on assessment resource availability.
17. TRBs shall be convened and conducted IAW with NPTORD 4500-54 "Training Review Board Process".
18. In the event of prolonged trainee absence, TE Staff shall determine, on a case by case basis, the necessity to convene a TRB.

RESPONSIBILITY FOR ASSESSMENT

19. Assessment of trainees is to be done IAW CFITES Vol. 7 "Evaluation of Learners".
20. NFS staff provide quality control by administering and scoring all ECs and PCs IAW the standards laid out in this QSP and the assessments developed by NTDC(P).

COURSE REPORTS

21. Course reports (CF 377) shall be prepared and distributed IAW DAOD 5031-9 "Course Reporting and Certificates" and amplified in NPTORD 4500-51 "Course Reports and Certifications".
22. Students will be graded pass or fail. The narrative will indicate the student's strengths and weaknesses based on PC results to provide information to the Career Manager.
23. The Course Report narrative comments could include:
    a. Ability to work with others;
    b. Ability to work under pressure;
    c. Written and verbal communication skills;
    d. Organizational skills; and
    e. Leadership ability.
24. Training limitations and / or incomplete training must be included in the course report.

CHAPTER 3 - PERFORMANCE AND ENABLING OBJECTIVES

AUTHORIZATION FOR THIS CHAPTER AND/OR AMENDMENTS MUST BE GRANTED BY THE COMMANDING OFFICER OF THE NAVAL TRAINING DEVELOPMENT CENTRE (PACIFIC)

PO 001 Navigate an Arctic and Offshore Patrol Vessel (AOPV) through Ice Covered waters

1. **Performance**: Navigate an AOPV through ice covered waters (T018)

2. **Conditions**:

   a. Given:

      (1) An AOPV;
      (2) Full bridge team;
      (3) Navigation aids;
      (4) Full chart outfit;
      (5) Access to ice products;
      (6) Access to meteorological products;
      (7) Regulatory documents;
      (8) Helicopter;
      (9) Unmanned Aircraft System (UAS);
      (10) CSTG risk assessment matrix;
      (11) Multi Role Rescue Boat (MRRB);
      (12) Support from CAF elements;
      (13) Assistance from OGDs;
      (14) Support from NGOs; and
      (15) References.

   b. Denied: Nil
   c. Environment:
      (1) Ice covered waters at any latitude;
      (2) Day/Night; and
      (3) At sea in an AOPV in any environmental condition.

3. **Standard**: The NWO shall navigate through ice covered waters (T018) IAW references found in Annex A, by:
   a. Planning navigation through ice covered waters (T001) by:
      (1) Determining mission feasibility, through :
      (2) Applying Maritime Protected Area (MPA) precautions in arctic areas of enhanced environmental sensitivity (T049);
      (3) Applying IMO standards for operations in remote areas (T045); and
      (4) Applying Arctic Shipping and Pollution Prevention Regulations (ASPPR) (T044);

(5) Identifying logistical and operational assets available in the Canadian Arctic (T013);
(6) Liaising with northern based OGDs and logistical assets (T036);
(7) Analysing equipment limitations in high latitudes, to include;
(a) Communications equipment and support; and
(b) Navigation equipment;

(1) Applying Canadian Ice Services products (T043) by:
(a) Interpreting information from Canadian Ice Services Sources (T005), by employing Arctic Ice Regime Shipping System (AIRSS);
(b) Determining ice regimes by calculating ice numerals (T008);
(c) Applying ECAREG and NORDREG recommended routing (T046); and
(d) Completing ice routing message (T011).

a. Manoeuvring ship in various ice concentrations and coverage (T020) by:
(1) Navigating the ship accounting for limitations in GPS electronic positioning systems (T015) by:
(a) Configuring navigation systems for northern latitudes (T017); and
(b) Determining safe depth of water by using Forward Looking Echo Sounder (FLES);
(2) Obtaining current ice information (T002) by:
(a) Measuring movement of sea ice (T003) to include:
i. Analyzing ice charts;
ii. Assessing course and speed made good in ice floes; and
iii. Assessing ice drift;
(b) Discriminating between radar targets and ice features in ice clutter with Sigma 6 Radar (T014)
(c) Performing Helicopter operations in northern latitudes (T032) to include:
i. Conducting helicopter ice recce; and
ii. Identifying and plotting dangers with a handheld GPS;
(3) Adjusting Navigation plan for dynamic ice conditions (T004) by:
(a) Preparing risk assessment before approaching ice (T019) to include:
i. AOPV engineering and torque considerations;
ii. Bollard tests;
iii. Damage control considerations; and
iv. Risk assessment matrix;

(b) Entering and exiting ice (T021);

(4) Navigating an AOPV through ice leads (T022) by:

(a) Employing techniques to avoid becoming beset in ice (T034) to include:

i. Utilizing ice control modes;

ii. Conducting a star manoeuvre; and

iii. Conducting backing and ramming procedures;

(b) Executing sit and drift procedures;

(c) Employing procedures to free own AOPV from ice by:

i. Utilizing heeling tanks; and

ii. Maintaining continuous forward propulsion;

(5) Provide assistance to a vessel beset in ice (T035);

(6) Conducting seamanship evolutions in ice (T031) to include:

(a) Refuelling an AOPV in the Arctic (T033);

(b) Executing berthing and slipping evolutions in ice covered waters (T023);

(c) Operating bow thruster while ship handling (T024);

(d) Anchoring in ice covered waters;

(e) Managing small boat operations in the Arctic; and

(f) Conducting towing operations in ice covered waters.

b. Manage operations with Canadian Coast Guard ice breaker (T026) to include;

(1) Conducting ice breaker escort operations by:

(a) Communicating with ice breaker; and

(b) Following ice breaker direction while moving in convoy;

(2) Protecting shore facilities from ice (T039);

(3) Performing harbour break out procedures (T040);

(4) Breaking up ice jams in channels affected by currents (T038) to include flood control procedures;

(5) Cutting and maintaining a channel in fast ice (T037); and

(6) Conducting pollution protection (clean up) in the Canadian Arctic (T050).

4. **Training Limitations**:

a. Simulator Arctic database;

b. Simulator AOPV platform modeling;

c. Simulator ice characteristics modeling;

d. Simulator AOPV and ice interactions modeling;

e. Difficulty in simulating environmental considerations; and

f. Human factors (eg. Fatigue, stress).

PO 001 Navigate an Arctic and Offshore Patrol Vessel (AOPV) through Ice Covered waters

EO 001.01 Plan navigation through ice covered waters

1. Performance: Perform a navigation appraisal.
2. Conditions:
   a. Given:
      (1) WECDIS;
      (2) Chart and Publication Outfit;
      (3) All CHS products
      (4) Access to Canadian Ice Services products
      (5) Access to meteorological products
      (6) Planning charts
      (7) Ship handling scenarios in ice
      (8) Access to ECAREG
      (9) Access to NORDREG
      (10) Access to MCOIN message drafter
      (11) NNF operating instructions;
      (12) Captain's ship's books;
      (13) Assistance; and
      (14) Instructor / Mentor; and
      (15) References.
   b. Denied: Nil;
   c. Environmental:
      (1) Classroom;
      (2) Day/Night; and
      (3) Collaborative environment.
3. Standard: The student shall plan navigation through ice covered waters (T001) by:
   a. Determining mission feasibility, by:
      (1) Integrating regulatory documents, by:

(a) Applying Maritime Protected Area (MPA) precautions in arctic areas of enhanced environmental sensitivity (T049);

(b) Applying IMO standards for operations in remote areas (T045);

(c) Applying Arctic Shipping Pollution Prevention Regulations (ASPPR) (T044);

(2) Identifying logistical and operational assets available in the Canadian Arctic (T013);

(3) Liaising with northern based OGDs and logistical assets (T036);

(4) Analysing equipment limitations in high latitudes, to include;

(a) Communications equipment and support; and

(b) Navigation equipment;

b. Applying Canadian Ice Services products (T043) by:

(1) Interpreting information from Canadian Ice Services Sources (T005), by employing Arctic Ice Regime Shipping System (AIRSS);

(2) Determining Ice regimes by calculating Ice numerals (T008);

(3) Applying ECAREG and NORDREG recommended routing (T046); and

(4) Completing Ice routing message (T011).

4. Time Required:

a. EO: 410 minutes; and

b. EC: 60 minutes.

5. Remarks: (limitations, development considerations)

a. Assistance is commensurate with what would be available from the bridge team and fellow navigating officers.

b. Refer to Annex F for clarification of terms.

c. Chart and publication outfit as issued to HMC Ships (as described in Annex F);

d. Run sheet as described in Annex F - Table 2.

6. **Instructional Methods**:

a. Pre-instruction package;

b. Interactive lecture;

c. Demonstration – performance; and

d. Self-study.

7. **Training Aids**:

a. Standard classroom configuration;

b. Lectures/presentations;

c. Training videos;

d. Mission scenario; and

e. References.

8. **Test Details**: EO 001.01 will be tested in EC 001.01 and PC 001.

9. **Remarks**:

a. Access to Captain's Ship's Book, through Commanding Officer of AOPV;

b. Students require access to pre-instruction package material prior to course start date; and

c. EC001.01 will be assigned day one; EC001.01 to be assessed immediately prior to PC.

10. **Supporting Knowledges:** K001, K002, K003, K004, K005, K006, K007, K017, K018, K019, K023 and K024

11. **Supporting Competencies**: Task Prioritization, Decision Making, Working Memory, Communication, Leadership, Motivation and Interpersonal Skill.

12. **Teaching Points**:

	Teaching Point Description	Instructional Method	Time (in min)	References	Comments
TP1	TP 1.1 Review of references and Documentation	Self-Study/Pre-Course reading package	A/R	A1, A2, A3, A5, A6, A14, A16, A44, A45, D1	Students to have reviewed all identified references prior to attending course. Portal / SharePoint with all of the references needs to be logically designed such that candidates are aware which references they should have read, and which are available for their convenience
TP2: Analyze mission direction : initial planning considerations					

	Teaching Point Description	Instructional Method	Time (in min)	References	Comments
TP2	SP 2.1 Introduce the Cyclical Planning Process	Interactive Lecture	20 min	A1	Identify organizations that contribute to the arctic planning process Assess Joint CONOPS Introduce the Arctic Marine Advisory Board and how they effect change in arctic charting
	SP 2.2 Apply Polar Code requirements (T045)	Interactive Lecture	30 min	B3, D1	IMO Polar Code Introduce ship classification systems IACS Equipment, design and construction Operations and manning Introduction to ice pilots in commercial industry Nuances between RCN and industry wrt certification Introduce fractal strength (incorporated in to PC and ice numerals)
	SP 2.3 Appraise AOPV ship design	Interactive lecture/ Self-Study	40 min	A1, A2, A4, A52, C1, C2, C6, C7, D3, D5	AOPV enhancements to include: Ice belt Ice breaker stern Ice strengthened propellers Ice knives Hull form Fin stabilizers Engineering Considerations: Propulsion arrangement, PGD, shaft line & torque fluctuations during ice breaking, propulsion control system to limit power – torque vs speed, ice control modes and heeling tanks

	Teaching Point Description	Instructional Method	Time (in min)	References	Comments
					Comparison to other ice capable ships, GC, Allied, Commercial Stability and parametric roll
	SP 2.4 Appraise AOPV ice capabilities	Interactive lecture	20min	A1, A2, A4, A52	Intended Ice Regime Milling and Stalling Trim Heeling Tanks Beaching on ice unintentionally
	SP 2.5 Apply ECAREG / NORDREG recommended routing (T046)	Interactive lecture/Demonstration	15 min	A14, A15, A46, A51	Apply to PC 5 ship Zone/Date system (T044) Introduction to ASSPR (when to use) Introduction to AIRSS (when to use) Discuss differences between CAC and PC classifications and when to use each
	SP 2.6 Analyze Marine Protected Areas (T049)	Interactive lecture	15 min	A14, A42, D6	Explain the nature of MPAs in North. Who determines an area is an MPA. How do you determine an area is an MPA? Pilot, chart symbol? Consider operational/environmental restrictions in MPA
	SP 2.7 Identify northern based logistical and operational assets. CAF ATC, NOH (T013)	Interactive lecture	30 min	A10, A11, A12, A13, A25, A27, A42, A44, A48, A49, A50, A51	Discuss liaising with JTFN Consider CAF ATC infrastructure Consider northern operation hubs and infrastructure Compare northern operations hubs to NNF

	Teaching Point Description	Instructional Method	Time (in min)	References	Comments
					Discuss liaising with Northern community leaders Identify best database/resource for finding CAF and other northern log assets. Something less static than RCN ops manual
	SP 2.8 Cultural/Aboriginal awareness training requirements	Interactive lecture	15 min	Open source information	Introduce the requirement for training for Ship's COY, and how to access on (DLN)
TP3: Mid planning considerations: Determining Routing					
TP3	SP 3.1 Identify 4-7 NWP routes	Interactive lecture	15 min	A18, A41, A42	Introduction to primary and secondary routing through NWP Discuss benefits / risks associated with each route
	SP 3.2 Identify Iceberg Ally	Interactive lecture	5 min	A14, A15, A47	Routing considerations in Davis Strait, Baffin Bay, and Labrador Sea Calving: primary source of icebergs
	SP 3.3 Identify seasonal variability of daylight	Interactive lecture	5 min	A14	Introduce graph in accordance with ref
	SP 3.4 Appraise Hamlets and available resources (T013)	Interactive lecture	20 min	A42, A43 Open source information	Identify special events Hamlet resource limitations Hamlet infrastructure limitations Staging supplies (think about overland /air access) Operational impact on Hamlet. Shore visits

	Teaching Point Description	Instructional Method	Time (in min)	References	Comments
					Visits and protocol planning Outreach to Hamlets
	SP 3.5 Identify medical support (T013)	Interactive lecture	10 min	A1, A25	Landing strips Hamlet medical support
	SP 3.6 Resupply in the Arctic (T013)	Interactive lecture	15 min	A25	Long lead time required Staging supplies Arctic resources through CCG, industry, Greenland (Nuuk)
	SP 3.7 Appraise fuel consumption in ice covered waters	Interactive lecture	10 min	A42, A43, A52	Fuel consumption in ice covered vs open ocean Distance tables
	SP 3.8 Account for fueling methods in the Arctic (T013)	Interactive lecture	20 min	A1, A2, A25, A27, A28	Fueling methods in the Arctic: NNF, CCG fittings and unique considerations Commercial industry fueling
	SP 3.9 Identify embarkation/disembarkation points	Interactive lecture	10 min	A1, A42, A43	CAF and OGD infrastructure
TP4 Final Planning Considerations					
TP4	SP 4.1 Apply Canadian Ice Services Products (T043)	Demonstration-performance EDI	30 min	A14, A15, A20, A21, A22, A23, B2	Instructor will demonstrate: Where to find products and how to source products Analyze information to assess impact on routing. Overlay techniques to determine movement of sea ice extent Introduce Ice-Vue and MODIS Worldview

	Teaching Point Description	Instructional Method	Time (in min)	References	Comments
	SP 4.2 Calculate ice numerals to determine ice regime (T008)	Demonstration	15 min	A14, A15, A20, A21, A22	Learned on Basic Arctic Navigation (BAN) course - refresher Tactical routing – plan at extent of ice floe
	SP 4.3 Apply CCG proven routing	Demonstration	10 min		Not based on ice, but rather to account for limited chart fidelity Account for Arctic environmental phenomena when planning navigation Identify rapid change in depth in arctic waters due to Pingos Identify areas of refuge along routing to include Polynyas
	SP 4.4 Tidal ranges	Interactive Lecture	10 min	A40, B2	Discuss unique tidal ranges in Arctic
	SP 4.5 Determine potential ice conditions	Demonstration - performance	10 min	A1, A2, A14, A15, A19, A20, A21, A22, B2	Adjust plan according to ice conditions
	SP 4.6 Analyze equipment limitations in high latitudes	Interactive Lecture	30 min	A1, A14, A41, A51	Chart surveys Datum shifts Communications limitations GMDSS limitations (WGS 84 datum) Limitations of small scale Mercator charts with distortions of distances and area
	SP 4.7 Complete ice routing message (T011)	Demonstration - performance	10 min	A14, A15	Students should be exposed to ice routing message, informed how RCN plans to participate Brief navigation plan, and present ice routing message

EC 001.01- Plan navigation through ice covered waters

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions: Classroom.

Arrangements and Procedures

1. EC 001.01 will be a practical assessment based on content from EO 001.01.

2. To pass this EC, the trainee must achieve SATs on all critical checklist items.

3. Time required: 60 min.

4. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.01 will be a practical assessment based on content from EO 001.01.

2. During this practical assessment, you will plan a navigation through ice covered waters. You must pass this EC to pass EO 001.01.

3. References will be provided.

4. Pass mark is SAT on all critical checklist items IAW ECL 001.01.

5. You will have 60 min to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Reference Annex K for EC/PC assessment plans.

PART IV – ENABLING CHECK LIST EC 001.01

Item	EC 001.01 Plan navigation through ice covered waters	SAT	UNSAT	Remarks
Note: Critical items are in ***bold/italics*** Note: If student is not required to complete a checklist item, mark as "N/A" in remarks column				
Analyse mission direction: Initial planning considerations				
Clearly defined mission, to include:				
Parameters		☐	☐	
Considerations		☐	☐	
Intentions		☐	☐	
Considered communications in plan to include the following: VHF, DSC, GMDSS, Weather broadcasts, mandatory/voluntary VTMS areas		☐	☐	
Examined areas of interest including:				
Environmental areas		☐	☐	
Geographic areas		☐	☐	
Mid Planning considerations: Determine Routing				
Selected appropriate route		☐	☐	
Calculated limitations to include:				
Speed		☐	☐	
Time		☐	☐	
Distance		☐	☐	
Designated hamlets are researched		☐	☐	
Completed fuel consumption calculations		☐	☐	
Identified refuelling plan		☐	☐	
Completed logistics resupply plan		☐	☐	
Final Planning Considerations				
Considered Navigation Hazards (such as Pingos, higher concentrations of ice and Icebergs, limited surveys, datum shifting)		☐	☐	

Calculated ice regime correctly	☐	☐	
Identified Tides, Tidal Streams and Ocean Currents as relates to plan	☐	☐	
Drafted Ice routing messages	☐	☐	

Score: SAT / UNSAT

Inability to calculate ice regime is considered critical, unsatisfactory calculations result in failure of EC.

Positive Feedback	Improvement Feedback

EC 001.01 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 001.01 with the candidate.	**Candidate** I have been debriefed on the result of EC 001.01.

EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 001.02 Manoeuvre Ship in Various Ice Concentrations and Coverages

1. **Performance**: Manoeuvre Ship in Various Ice Concentrations and Coverages (T020).

2. **Conditions**:

   a. Given:

      (1) WECDIS;
      (2) Full chart outfit;
      (3) All CHS products;
      (4) Access to Canadian Ice Services Products;
      (5) Access to meteorological products;
      (6) Planning charts;
      (7) Ship handling in ice scenarios;
      (8) Access to ECAREG;
      (9) Access to NORDREG;
      (10) NNF operating instruction;
      (11) Captain's Ships Books;
      (12) Assistance;
      (13) Helmsman;
      (14) Instructor / Mentor;
      (15) NABS;
      (16) Unique Ice Services Canada Software;
      (17) Ice Radar (Sigma 6);
      (18) FLES;
      (19) Supervision;
      (20) Risk assessment matrix; and
      (21) References.

   b. Denied: Nil.

   c. Environment:

      (1) Classroom;
      (2) Simulator;
      (3) Day/Night; and
      (4) Collaborative environment.

3. **Standard**: The student shall manoeuvre an AOPV in various ice concentrations and coverages (T020) by:

a. Navigating the ship accounting for limitations in GPS electronic positioning systems (T015) by:

(1) Configuring navigation systems for northern latitudes (T017); and

(2) Determining safe depth of water using the Forward Looking Echo Sounder (FLES);

b. Obtaining current ice information (T002) by:

(1) Measuring movement of sea ice (T003) to include:

(a) Analyzing ice charts;
(b) Assessing Course and Speed made good in ice floes; and
(c) Assessing Ice drift;

(2) Discriminating between radar targets and ice features in ice clutter with Sigma 6 Radar (T014);

(3) Performing Helicopter operations in northern latitudes (T032) to include:

(a) Conducting helicopter ice recce; and
(b) Identifying and plotting dangers with a handheld GPS;

c. Adjusting Navigation plan for dynamic ice conditions (T004) by:

(1) Preparing risk assessment before approaching ice (T019) to include:

(a) AOPV engineering and torque considerations;
(b) Bollard tests;
(c) Damage control considerations; and
(d) Risk assessment matrix;

(2) Entering and exiting ice (T021);

d. Navigating an AOPV through ice leads (T022) by:

(1) Employing techniques to avoid being beset in ice (T034) to include:

(a) Utilizing ice control modes;
(b) Conducting a star manoeuvre; and
(c) Conducting backing and ramming procedures;

(2) Executing sit and drift procedures;

(3) Employing procedures to free own AOPV from ice by:

(a) Utilizing heeling tanks; and
(b) Maintaining continuous forward propulsion;

e. Freeing an independent vessel beset in ice (T035);

f. Conducting seamanship evolutions in ice (T031) to include:

(1) Refuelling an AOPV in the Arctic (T033);

(2) Executing berthing and slipping evolutions in ice covered waters (T023);

(3) Operating bow thruster while ship handling (T024);

(4) Anchoring in ice covered waters;

(5) Managing small boat operations in the Arctic; and

(6) Conducting towing operations in ice covered waters.

4. **Time Required**:

a. EO – 810 minutes; and

b. EC – 150 minutes (3 phases at 50 minutes each).

5. **Instructional Methods**:

a. EDI;

b. Interactive Lecture; and

c. Case Study.

6. **Training Aids**:

a. Standard classroom configuration;

b. Lectures/Presentations;

c. Training videos;

d. Case Study;

e. NABS; and

f. References.

7. **Test Details**: EO 001.02 will be tested in EC 001.02 and PC 001.

8. **Remarks:**

a. Access to Captain's ships book, through Commanding Officer of AOPV; and

b. Case study for TP 6.7 to be made available to students during day one during course intro.

9. **Supporting Knowledges:** K008, K009, K010, K011, K012, K013, K014, and K015.

10. **Supporting Competencies:** Situational Awareness, Spatial Orientation, Task Prioritization, Divided Attention, Decision Making, Long Term Memory (LTM), Working Memory (WM), Stress Tolerance, Conscientiousness, Communication, Leadership, Motivation and Interpersonal Skills.

11. **Teaching Points:**

	Teaching Point Description	**Instructional Method**	**Time (in min)**	References	Comments
TP1: Navigate the Ship Accounting for Limitations in GPS and electronic positioning systems (T015)					
TP1	SP 1.1 Configure navigation systems for northern latitudes (T017)	EDI	50 min	A1, A4, A14, B1, C3, C4, C5	GPS positions vs PI Datum configuration Ensuring that GMDSS datum is adjusted to that of chart Charts: Projection of chart in use Accuracy in surveys

	SP 1.2 Tuning the Radar	EDI	20 min	A1, A4, A14, B1, C5	Detuning radar to enhance RACON return Differentiating echo images in various ice conditions Differentiating between shoreline and ice Limited radar cross sections of ice floes
	SP 1.3 Determine safe depth of water using FLES	EDI	20 min	C4	Techniques to employ FLES Bottom topology change due to ice movement Bummocks using FLES To determine iceberg keels
TP2: Obtain current ice information (T002)					
TP2	SP 2.1 Measure movement of sea ice (T003)	EDI	15 min	A4, A14, A15	Overlaying CIS charts and radarsat imagery, aircraft reconnaissance reports 24-48-72 hour ice charts Ice Vue program vs. hand drawn Identify stages of ice decay and development
	SP 2.2 Discriminate between ice features using radar (T014)	EDI	20 min	A14, C5	Identify ice features in ice clutter Hummock Detune radar for RACON detection Appropriate range scale Detect ice bergs

	SP 2.3 Perform helicopter operations to conduct ice recce (T032)	EDI/Video	20 min	A1, A14	Consider who to send up in helicopter Plotting route in hand held GPS, or other plotting tool to determine the path of least resistance, and areas of concern (ridging) Photography and video equipment
	SP 2.4 Determining path of least resistance	EDI/Video	20 min	A1, A2, A14, A21, A22	Ice characteristics Stages of decay and development
	SP 2.5 Reporting current ice conditions	Interactive Lecture	15 min	A5, A14, A15	Providing feedback to METOC or directly to CIS of actual ice conditions
TP3: Adjust navigation plan for dynamic ice conditions (T004)					
TP3	SP 3.1 Enter and exit ice (T021)	EDI	50 min	A1, A2, A4, A14	Identify time to get through the ice floe Adjust waypoints based on path of least resistance Angle of incident Speed of entrance: Speed vs. power Equation to determine impact Beaching: Speed vs weight Weight vs efficiency through ice DC considerations Determine safe distance from icebergs: Calculate keel based on height

					Determine stability of icebergs based on stages of decay
	SP 3.2 Prepare risk assessment before approaching ice (T019)	EDI	15 min	A2	Risk assessment matrix: Conducting Bollard Test Results of Bollard Test Engineering Considerations: Time/torque limits Time at above 100% torque before back away and ram Ice modes Power management Division of duties: Helm Throttles Conn Torque Bridge Brief
	SP 3.3 Navigate at night in ice covered waters	EDI	20 min	A2, A4, A14	Setting up lighting (search lights) Safe speed (backscatter)
TP4: Navigate an AOPV through ice leads (T022)					
TP4	SP 4.1 Employ techniques to avoid becoming beset in ice (T034)	EDI	50 min	A1,A14,C7, D3, D5	Defining ice pressure Impact of ice pressure (fractal strength) Select ice control modes Monitoring MLA and speed Ramming techniques: Management of speed vs power vs torque Significance of shadow zone and power Backing techniques: Rudder configuration

					Propeller wash to clear astern Distance Star manoeuvre Hull Friction: Reducing via Bubbler system/firehose, Anchor wash
	SP 4.2 Navigate in coastal waters amongst ice floes	EDI	25 min	A1, A14, A34, A42	Avoiding being forced aground by ice floes Impact of interacting with fast ice: Define fast ice Shoreline infrastructure Inside/Outside passage in Western Arctic Running with and stemming a current in the Arctic
	SP 4.3 Execute Sit and Drift procedures	Interactive lecture	20 min	A1, A2, A14, B3	Coming to a stop: Ice floe drift (rate and direction) Assessing the ship in ice floe Considerations: Time of day Hazards of night navigation (search lights) Ice characteristics (combination of first, second and multi-year ice) Crew rest Bridge team experience Instructions to OOW Propulsion configuration (DGs) Proximity to the PIM Propeller considerations (avoiding freeze-up and damage to propellers by ice)

TP4	SP 4.4 Employ procedures to free an AOPV from ice	EDI	30 min	A2, A14	Propeller and rudder considerations Propulsion configuration (DGs) requirement for full power Reducing friction and pressure on the hull Use of heeling tanks: Characteristics of heeling tanks Effects on hull, personnel and equipment Request for assistance from another Icebreaker
TP5: Provide assistance to a vessel beset in ice (T035)					
TP5	SP 5.1 Evaluate tasking to provide assistance	Interactive lecture	20 min	A2, A14	Tasking received from RJOC/MCC or vessel to vessel Considerations: Ice regime Environmental conditions Circumstance of besetment Vessel characteristics Determine feasibility: Fortitude to say "no"
	SP 5.2 Render assistance to vessel beset in ice	EDI	15 min + 100 min	A2, A14, A29	Communicate intentions with vessel: Plan Requirements of beset vessel Language barriers Reduce ice pressure on beset vessel Breakout procedures: Cut-out procedure: Create an ice fender (preferable on both sides) Relieve pressure

					Position AOPV 5 ca ahead of vessel on third approach Begin escort Back-down procedure (no requirement to relieve pressure on vessel): Vessel does not require cut-out procedure Using astern propulsion creating path for beset vessel to safely navigate Begin escort
	SP 5.3 Take-up escort	EDI	15 min	A1, A2, A14, A29	Safe distances: Minimum distance based on escorted vessel, speed and ice conditions Experiment with power levels Considerations of shouldering: Conditions of Ice; Beam and maneuverability of escorted ship Competence of AOPV helmsman Dual escort
	SP 5.4 Breaking off the escort	EDI	10 min	A1, A2, A14, A29	Environmental conditions and Ice regime Conditions and capabilities of the escorted vessel Determine routing of escorted vessel

TP6: Conduct seamanship evolutions in ice covered waters (T031)					
TP6	SP 6.1 Refuel an AOPV in the Arctic (T033)	EDI	15 min 50 min	A1, A2, A25, A27	NNF: NNF infrastructure Fueling alongside position: Placement of lines Contour of ship vs jetty cells Fuel tower assembly Topology of fueling positions Secondary fueling by hose Environmental clean-up capacity and responsibilities between contractor and HMC ship CCG: Fueling alongside ship at anchor: CCG ship characteristics: Fuel position and anchor Assessing CCG ship yaw, reducing using bubbler Conducting approach: CCG assistance in making approach Appropriate fuel connections CCG are the lead in pollution clean-up in Canadian Domestic Waters Commercial: Disposition of anchor pattern Receiving barge while at anchor
	SP 6.2 Execute berthing procedures in ice covered waters (T023)	EDI	15min 50 min	A1, A2, A4, B1	Approach considerations: Jetty fouled vs clear Jetty characteristics Ice characteristics (brash vs fast)

					Environmental (current, wind and moving ice) Clearing the alongside position CCG assistance (fast ice) vs AOPV (brash ice) Liability for damages Hazards: Brash vs fast ice Passing through the shear line Jetty characteristics Ice interaction with shore infrastructure & vessels in close proximity Approach techniques: Standard approach Using a spring line to hold ship's head to clear alongside position
	SP 6.3 Execute slipping procedures in ice covered waters (T023)	EDI	15 min	A1, A2, A4, B1	Departure considerations: Ice pressure (hull, lines and ice flow) Ice accretion on upper decks – urea / fertilizer Mitigate build-up (rudder and props) Adjusting Trim Stern tube water flow techniques Turning shaft by turning gear Use of heeling tanks to relieve pressure INFP (engines before lines) Seamanship (line handling) Proceeding astern: Use of rudder Flushing the stern (of ice)- Bow thruster Props Introducing the stern into ice

					Departures: fwd vs astern departure (based on ice conditions, current and moving ice)
	SP 6.4 Operate Bow thrusters while ship handling (T024)	EDI	15 min	A4, B1, A52	Moving athwart ships in ice: Bow thruster bollard push/pull Ice build-up against ship's hull Depth of thruster in relation to ice build-up Bow thruster makeup: Protective screen Direction of thruster rotation
	SP 6.5 Conduct anchorage in ice covered waters	EDI	20 min	A1, A2, A3, A4, A14, A42, A52, B1	Techniques: Conventional Deadman's byte In iceberg fields Considerations: Ice accretion on anchor cable, hawse pipe and ground tackle Length of cable – short stay Maintain power to windlass Bollard pull of windlass Density of ice (brash, rotten and scattered floes) Position of ice pack/floe Engineering configuration (NFP) Anchor watch (manning and use of RADAR) Potential requirement to expedite weighing anchor Precautions: Avoid dragging anchor Avoid getting forced against fast ice (ice floes) Avoid icebergs and calving fields
	SP 6.6 Manage small boat operations	Interactive Lecture	40 min	A1, A2, A3, A4, A10	MRRB: Capabilities: Nav suite Endurance

					Considerations: Depth of water in arctic Observing tidal cycles Charting and surveys Northern bottom topography Concentration of ice Time/distance/contingency planning for tasking Away team concept/construct Changing environmental conditions Crew payload – Arctic PPE Self-defense weaponry (for wildlife encounters) Comms equipment Survival equipment Helo/UAS availability Operations: OTH Beach operations Personnel and cargo transfers
	SP 6.7 Conduct towing operations in ice covered waters	Interactive Lecture/Case study	40 min	A1, A2, A4, A14, A52, B1	Types of vessels capable of towing in ice Occasions: Seriously disabled Short term Considerations: Ice regime (pancakes ice up to 6/10) Short line Long line Bridal Circumstance of stricken vessel: Power Steerage Ice classifications Hazards/Risks:

					Ice pressure on stricken vessel Striking ice AOPV towing capabilities (design) Yawing of the towed vessel

EC 001.02 - Manoeuvre Ship in Various Ice Concentrations and Coverages

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. NABS/Classroom.

Arrangements and Procedures

1. EC 001.02 will be a practical assessment based on content from EO 001.02.

2. To pass this EC, the trainee must achieve SATs on all critical checklist items.

3. Time required: 150 min.

4. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.02 will be a practical assessment broken into three phases, based on content from EO 001.02.

2. During this practical assessment, you will manoeuvre a ship in various ice concentrations and coverages. You must pass this EC to pass EO 001.02.

3. References will be provided.

4. Pass mark: SAT on all critical checklist items IAW ECL 001.02.

5. You will have 50 min to complete each phase of the assessment for a total of 150 min.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Reference Annex K for EC/PC assessment plans.

PART IV – ENABLING CHECK LIST EC 001.02

Item	EC 001.02 Manoeuvre Ship in Various Ice Concentrations and Coverages	SAT	UNSAT	Remarks
Note: Critical items are in ***bold/italics*** Note: If student is not required to complete a checklist item, mark as "N/A" in remarks column				
Phase 1: Navigate in coastal waters, entering/exiting ice, navigating through ice leads				
	Selected appropriate route	☐	☐	
	Delineated roles and responsibilities of Bridge Team	☐	☐	
	Introduced Hull into Ice accounting for:			
	Speed	☐	☐	
	Power	☐	☐	
	Torque	☐	☐	
	Angle of incident	☐	☐	
	Managed precautions to avoid becoming beset in ice as required:			
	Maintained a safe speed	☐	☐	
	Applied proper backing and ramming techniques	☐	☐	
	Conducted Star Maneuver	☐	☐	
	Adapted to environmental conditions/visibility to include as required:			
	Search lights	☐	☐	
	Speed	☐	☐	
	Deployment of helicopter	☐	☐	
	Augmenting bridge team	☐	☐	
	Go/No-Go	☐	☐	
Phase 2: Freeing a beset vessel from ice / escort operations				
	Established communication with beset vessel	☐	☐	
	Communicated intentions with beset vessel	☐	☐	

Conducted cut out procedure to reduce pressure on beset vessel	☐	☐	
Provided direction for escorted vessel to include:			
Maintaining safe distance	☐	☐	
Speed and speed changes	☐	☐	
Emergency signals and procedures	☐	☐	
Breaking off escort	☐	☐	
Phase 3: Proceed alongside jetty fouled by ice			
Adapted approach based on ice characteristics	☐	☐	
Adapted approach based on jetty characteristics	☐	☐	
Cleared alongside position of ice:	☐	☐	
Proceeded alongside:	☐	☐	

Score: SAT / UNSAT

Positive Feedback	Improvement Feedback
Positive Feedback	**Improvement Feedback**

EC 001.02 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 002 with the candidate.		**Candidate** I have been debriefed on the result of EC 002.	
PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 001.03 Manage Operations with CCG Icebreakers

1. **Performance**: Manage Operations with CCG Icebreakers. (T026).

2. **Conditions**:

   a. Given:

      (1) CCG guest speaker;
      (2) Access to Canadian Ice Services products;
      (3) Planning charts;
      (4) Instructor; and
      (5) References.

   b. Denied: Nil

   c. Environment:

      (1) Classroom;
      (2) Simulator;
      (3) Day/Night; and
      (4) Collaborative environment.

3. **Standard**: The student shall manage operations with Canadian Coast Guard Ice breaker (T026) to include:

   a. Conducting ice breaker escort operations by:

      (1) Communicating with ice breaker; and
      (2) Following ice breaker direction while moving in convoy;

   b. Protecting shore facilities from ice (T039);

   c. Performing harbour break out procedures (T040);

   d. Breaking up ice jams in channels effected by currents (T038) to include:

      (1) Flood control procedures;
      (2) Cutting and maintaining a channel in fast ice (T037); and
      (3) Conducting pollution protection (clean up) in the Canadian Arctic (T050).

4. **Time Required**:

   a. EO – 170 minutes; and

   b. PC – 150 minutes.

5. **Instructional Methods**:
   a. Interactive lecture; and
   b. Guest speaker.

6. **Training Aids**:
   a. Standard classroom configuration;
   b. Lectures/Presentations;
   c. Training videos;
   d. Case Study;
   e. NABS; and
   f. References.

7. **Test Details**: EO 001.03 will be tested in PC 001.

8. **Remarks**: Based on availability of CCG guest speaker, consider video teleconference.

9. **Supporting Knowledges:** K020, K021, K022 and K025**.**

10. **Supporting Competencies:** Long term memory (LTM), Communication, Motivation, and Interpersonal Skills.

11. **Teaching Points**:

	Teaching Point Description	Instructional Method	Time (in min)	References	Comments
TP1: Conduct Ice Breaker Escort Operations					
TP1	1.1 Communicate with an Icebreaker	Interactive Lecture/Guest speaker	10 min	A14, A23, A29	Methods: Sound signals VHF Interco Emergency signals Info required by the CCG Icebreaker Captain Icebreaker instructions Anticipated responses
	1.2 Prepare for escort	Interactive Lecture/Guest speaker	10 min	A14, A23, A29	Review latest forecast and ice information Freeing a beset vessel
	1.3 Conduct escort	Interactive Lecture/Guest speaker	10 min	A14, A23, A29	Safe speed Escort distance: Ice conditions Ice pressure Stopping distance of escort Visibility Escorting vessel with wide beam BRM during escort Convoy operations
TP 2: Protect shore facilities from ice					

	Teaching Point Description	Instructional Method	Time (in min)	References	Comments
TP2	2.1 Protect shore facilities from ice	Interactive Lecture/Guest speaker	15min	A14, A15, A23, A29	Considerations Preparations Techniques
TP3: Preform harbor break out procedures					
TP3	3.1 Conduct harbor break out procedures	Interactive Lecture/Guest speaker	15min	A14, A23, A30, A31	Considerations Preparations Techniques
TP4: Break up ice jams in channels effected by current					
TP4	4.1 Break up ice jams in channels effected by current	Interactive Lecture/Guest speaker	30min	A14, A23, A29, A30	Considerations Preparations Techniques
TP5: Cut a channel in fast ice					
TP5	5.1 Cut a channel in fast ice	Interactive Lecture/Guest speaker	30min	A14, A23	Considerations Preparations Techniques
TP6: Conduct pollution protection in the Canadian Arctic					
TP6	6.1 Conduct pollution protection in the Canadian Arctic	Interactive Lecture/Guest speaker	25min	A14, A23	Considerations Preparations Techniques
TP7: CCG Operations in the Arctic					
TP7	7.1 C2 Structure of the CCG	Guided Discussion/ Guest Speaker	25 min	A14, A15, A23	RCN participants prepare to compare CCG C2 to RCN C2 structure.

	Teaching Point Description	Instructional Method	Time (in min)	References	Comments
					CCG to discuss previous year's arctic operations

PC 001 Navigate an AOPV through ice covered waters

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions: NABS/Classroom

Arrangements and Procedures

1. PC 001 will be a practical assessment based on content from PO 001.

2. To pass this PC, the trainee must achieve SATs on all critical checklist items.

3. Time required: 120 min (60 for initial PC, 60 for retest/remedial).

4. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 001 will be a practical assessment based on content from PO 001.

2. During this practical assessment, you will navigate an AOPV through ice covered waters. You will be provided with a a specific scenario relating to ice breaking. You must pass this PC to pass PO 001.

3. References will be provided.

4. Pass mark SATs on all critical checklist items IAW PCL 001.

5. You will have 60 min to complete this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

Reference Annex K for EC/PC assessment plans.

PART IV – PERFORMANCE CHECK LIST PC 001

Item	PC 001 Navigate an AOPV through Ice covered waters	SAT	UNSAT	Remarks
Note: Critical items are in ***bold/italics*** Note: If student is not required to complete a checklist item, mark as "N/A" in the remarks column				
Calculated ice regime correctly		☐	☐	
Identified Tides, Tidal Streams and Ocean Currents as relates to plan		☐	☐	
Selected appropriate route		☐	☐	
Delineated roles and responsibilities of Bridge Team		☐	☐	
Introduced hull into ice accounting for:				
Speed		☐	☐	
Power		☐	☐	
Torque		☐	☐	
Angle of incident		☐	☐	
Managed precautions to avoid becoming beset in ice as required:				
Maintained a safe speed		☐	☐	
Applied proper backing and ramming techniques		☐	☐	
Conducted Star Manoeuvre		☐	☐	
Adapted to environmental conditions/visibility to include:				
Speed		☐	☐	
Go/No-Go		☐	☐	
Search lights		☐	☐	
Deployment of helicopter		☐	☐	
Augmenting bridge team		☐	☐	
Established Communication with beset vessel		☐	☐	
Communicated intentions with beset vessel		☐	☐	

Conducted cut out procedure to reduce pressure on beset vessel	☐	☐	
Provided direction for escorted vessel to include:			
Maintaining safe distance	☐	☐	
Speed and speed changes	☐	☐	
Emergency signals and procedures	☐	☐	
Breaking off escort	☐	☐	
Adapted approach based on ice characteristics	☐	☐	
Adapted approach based on jetty characteristics	☐	☐	
Cleared alongside position of ice	☐	☐	
Proceeded alongside	☐	☐	

Score: SAT / UNSAT

Unsat on any critical item results in failure of PC.

Positive Feedback	Improvement Feedback

PC 001 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of PC 001 with the candidate.		**Candidate** I have been debriefed on the result of PC 001.	
PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

Annex A – Main References

(**Annex A not to be deleted**)

A = CAF or other official Canadian government publication
B = Official Allied Force or Allied government publication
C = Commercial publication
D = Other

Ref Code	Publication
A1	RCN Arctic Operations Manual
A2	Command of an Arctic Offshore Patrol Vessel (AOPV)
A3	Harry DeWolf Class AOPS – CONUSE ver. 1.0
A4	CFCD 130 Canadian Navigation Manual ver.1.7
A5	MANICE Canadian Ice Services
A6	CJOC Plan for the North
A7	Arctic Waters Pollution Prevention Act
A8	CJOC Commander's Planning Guidance – Operation NANOOK 2018
A9	Operation NANOOK 2018 – Maritime Exercise Initial Planning Conference Administrative Instructions
A10	Canadian Forces Northern Employment and Support Plan, Nov. 2012
A11	Optimal RSOM-hub Locations for Northern Operations (Defence Research and Development Canada)
A12	MOU Between DND and DFO Concerning training of RCN personnel on Ships of the CCG operating North of 60 degrees
A13	MOU Between DFO/CHS) and DND regarding provision and distribution of hydrographic charts and publications, data, and mutual cooperation
A14	Ice Navigation in Canadian Waters (CCG)

Ref Code	Publication
A15	Canadian Ice Services (https://www.canada.ca/en/environment-climate-change/services/ice-forecasts-observations/latest-conditions.html)
A16	Northern Approaches – The Army Arctic Concept 2021
A17	Polar Continental Shelf Program – Arctic Operations Manual
A18	The Canadian Maritime Arctic and Northwest Passage
A19	Royal Canadian Navy Intelligence, Surveillance, Target Acquisition and Reconnaissance Unmanned Aircraft System (RCN ISTAR UAS) Project
A20	TP12259E Arctic Ice Regime Shipping System (AIRSS) Standards
A21	TP12819E User Assistance Package for the Implementation of Canada's Arctic Ice Regime Shipping System (AIRSS)
A22	TP14044E Arctic Ice Regime Shipping System Pictorial Guide
A23	TP4338E Command of an Ice Breaker Volume 3
A24	Tri Command Arctic Action Plan
A25	Arctic Operations and Northern Re-Supply
A26	Statement on Canada's Arctic Foreign Policy
A27	Nanisivik Naval Facility – Concept of Support to Operations
A28	HDW Class SEMS
A29	Ice Breaking Operations Directive 1: Provision of Ice Breaking Services (DFO/CCG)
A30	Ice Breaking Operations Directive 2: Commercial Harbour Breakouts (DFO/CCG)
A31	Ice Breaking Operations Directive 3: Fishing Harbour Breakouts(DFO/CCG)

Ref Code	Publication
A32	Ice Breaking Operations Directive 4: Support to Sealing Vessels (DFO/CCG)
A33	Hydrographic Services Handbook (HSO)
A34	Notices to Shipping – Central Canada and the Arctic
A35	Radio Aids to Marine Navigation – Atlantic, St. Lawrence, Great Lakes, Lake Winnipeg and Arctic
A36	Radio Aids to Marine Navigation – Pacific and Arctic
A37	Annual Edition of Notices to Mariners – Central and Arctic
A38	List of Lights, Buoys and Fog Signals (Vol. 1-4)
A39	The Canadian Aids to Navigation System
A40	CHS Canadian Tide and Current Tables (Vol. 1-4)
A41	CHS Nautical Charts and Publications (Catalogue 4 – Arctic)
A42	CHS Northern Canada Sailing Directions (ARC 400E – ARC 404E)
A43	CHS Atlantic Coast Sailing Directions (ATL 100E – ATL 121E)
A44	DAOD 8007-0 –Notification of DND and CF Activities Within the JTF(N) AOR
A45	MARLANT Standing Environmental Assessment for Arctic Operations
A46	Marine Communications and Traffic Services (MCTS) – Central & Arctic Region
A47	CIS Annual Arctic Ice Atlas
A48	MOU Between DND and DFO Concerning Fisheries Patrols and Surveillance
A49	MOU Between DND and Denmark Concerning Arctic Cooperation
A50	MOU Between DND and DFO Concerning HMC Ship Personnel as Fisheries Officer

Ref Code	Publication
A51	Arctic Voyage Planning Guide
A52	HMCS Harry DeWolf – Captain's Ship Book
A53	Mariner's Handbook
B1	BR45(6) – Ice Navigation
B2	MODIS Worldview Imagery (https://coastwatch.glerl.noaa.gov/modis/)
B3	ATP 17 Chapter 10
C1	Ice Classed Ships: Main Engines (MAN)
C2	M303 – Arctic Offshore Patrol Ship (Odense Maritime Technology)
C3	WECDIS Operator's Manual
C4	FLES Operator's Manual
C5	Sigma 6 Operator's Manual
C6	Lloyd's Standard for Polar Class Ships
C7	American Shipbuilding Standards
D1	IMO Polar Code 01 January 2017
D2	IMO Guidelines for Ships Operating in Polar Waters
D3	Ice-Induced Loads on Ship Hulls (Norwegian University of Science and Technology)
D4	Arctic Marine Operations Report
D5	IACS Requirements Concerning Polar Class
D6	MARPOL – International Convention for the Prevention of Pollution from Ships Lloyd's Register Version 9.4

Annex B – Task List

(**Annex B not to be deleted**)

Includes train/no train decisions, priority and substantiation

Task #	Task Statement	T/NT/ Priority	Substantiation
Voyage Planning and reporting			
T001	Plan navigation through ice covered waters	T / M	
T002	Obtain current ice information	T/M	Advanced Techniques
T003	Measure movement of sea ice based on wind/set, charts	T/M	Advanced Techniques, ensure CMG, SMG
T004	Adjust navigation plan for dynamic ice conditions	T / M	
T005	Interpret information from Canadian Ice Services sources	T / L	
T006	Operate AOPV navigation systems	NT	Trained on FNO, separated points for Sigma 6 / FLES which need to be trained
	Administer chart outfit	NT	Trained on FNO
T007	Administer AOPV communications systems	NT	Trained on ICT
T008	Calculate Ice Numerals	T / L	Introduced on BAN, application on AAN
T009	Report ice observations	NT	Should be incorporated in basic arctic ops course
T010	Appraise impact of artic meteorological and weather conditions on passage plan	NT	Exposed to MET conditions on BAN, FNO, CDC
T011	Complete ice routing message	T / L	
T012	Transmit after action reports after passage in northern latitudes	NT	Function of OPS
T013	Identify logistical and operational assets available in the Canadian Artic	T / L	Delta training so CO XO NAV O don't take AOC
Equipment limitations			
T014	Discriminate between radar targets and ice features in ice clutter	T / M	
T015	Navigate the ship accounting for limitations in GPS of electronic positioning systems	T / M	
T016	Evaluate high latitude errors in compass bearings	NT	Trained on BAN

Task #	Task Statement	T/NT/ Priority	Substantiation
T017	Configure navigation systems for northern latitudes	T / L	
Operate and Manoeuvre ship in ice			
T018	Navigate an AOPV through ice covered waters	T / H	
T019	Prepare risk assessment before approaching ice	T / L	
T020	Manoeuvre ship in various ice concentrations and coverage	T / M	
T021	Enter and exit ice	T / M	
T022	Navigate an AOPV through ice leads	T / M	
T023	Execute berthing and slipping evolutions in ice covered waters	T / M	
T024	Operate bow thruster while ship-handling	T / M	
T025	Collaborate with MCTS in Canadian Arctic	NT	Already trained on NWO 3 & 4
T026	Manage operations with CCG icebreaker	T/H	Advanced Topics
T027	Communicate with icebreaker	NT	Already trained on BAN
T028	Request icebreaker support	NT	Already trained on BAN
T029	React to emergencies in ice	NT	Covered at NOPQ level
T030	Perform flood control in northern latitudes	NT	Knowledge component of T038
T031	Conduct seamanship evolutions in ice	T / M	
T032	Perform helicopter operations in northern latitudes	T / M	More than just landing and recovering
T033	Refuel AOPV in Arctic	T / L	NNF
T034	~~Manage precautions to~~ Employ techniques to avoid becoming beset in ice	T / M	
	Employ procedures to free vessel from ice	T / M	STAR Manoeuvre, Heeling tanks
T035	Provide assistance to a vessel beset in ice	T / H	
T036	Liaise with northern based OGDs and logistical assets	T / M	Delta training so CO XO NAV O don't take AOC
T037	Cut a channel in fast ice	T/H	Advanced topics for T26
T038	Break up ice jams in channels affected by current	T/H	Advanced topics for T26
T039	Protect shore facilities from ice	T/H	
T040	Perform harbour breakout procedures	T/H	Advanced Topic for T26
T041	Perform harbour breakout procedures in fishing ports	NT	Redundant with T040

Task #	Task Statement	T/NT/ Priority	Substantiation
T042	Submit invoices to clients regarding cost recovery actions	NT	Be covered in policy or OPORD
Regulatory documents			
T043	Apply Canadian Ice Services products	T/M	
T044	Apply Arctic Shipping pollution prevention regulations	T/L	
T045	Apply IMO standards for operation in remote area	T/L	
T046	Apply ECAREG or NORDREG recommended routing	T/L	
T047	Submit arctic operations reports	NT	Be covered in policy or OPORD
T048	Submit recommendations on aids to navigation in the Canadian Arctic	NT	Be covered in policy or OPORD
Environmental Factors and Regulations			
T049	Apply precautions in arctic areas of enhanced environmental sensitivity	T/M	
T050	Conduct pollution protection (clean up) in the Canadian Arctic	T/H	

Annex C –Record of Proceedings QSP Working Group
(**Annex C not to be deleted**)

Dates: 31 Jan – 14 Feb 2018

Location: HMCS Harry De Wolf Shore Office

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Gleason	C.L.E.	Cdr	00207	HDW
Ronaldson	M.G.	Lt(N)	00207	NTDC(P)
Zeeman	L.B.	Lt(N)	00207	NTDC(P)
Stevens	T.C.	Lt(N)	00207	NFR
Powell	L.R.	Lt	00211	NTDC(A)

Working Group Objective:

To create review and finalize the QSP for the advanced arctic navigation course.

Discussions / Decisions:

Prior to convening formally in person board members were presented with a series of references contained here-in, notably "Command of an AOPV" provided by the chair that was consulted throughout the board process.

Significant points of discussion were as follows:

Discussion Point 1:

Senior Met tech(s), where do they fit in to the series of courses? After a number of discussions throughout the board process it was determined that the Advanced Ice Navigation Course, due to the focus on command decisions and ship handling, was not the place to include Met personnel. The basic principles of ice navigation and meteorology needed to be introduced at an early stage in the basic arctic navigation course (to be developed).

Consideration for inclusion of Met techs on the basic arctic navigation course should be discussed during the convening of the BAN QSP board. Additionally, the board determined that since there were large areas where there would not be Met specific training, the Met occupation could consider training ice observations, reporting and arctic operations during their own coursing.

Since this is a course for CO, XO and NavO, it was determined that there would be no need for Met tech inclusion at this level.

Any new equipment will be trained by the contractor. Met techs would be aboard an AOPV in an ice observer role, not forecasting ice.

Basic Arctic Navigation review board will include discussion regarding a module for met techs.

Discussion Point 2:

Target population analysis:

Who has the most/least experience, not all CO/XO will have FNO experience? Also, how do reserves fit into the equation?

Despite the USQ indicating that Reserves can partake in the course and potentially fill positions aboard an AOPV, the lack of an NOP qualification for reserves is concerning. There is no formal mechanism whereby the RCN can guarantee reserve force officers have been exposed to Helo Ops, Replenishments at Sea etc. The most relevant reserve qualification identified was the OOW (MITE code granted), but again, this is not standardized. The AAN course is designed to teach individuals with major warship experience, lacking exposure to ice. Many skills gained aboard a major warship are considered transferabe, therefore, there is little to no review of "routine" operations. This is not the place for delta training for reserve officers. In order to take the course you must have attained NOPQ (OFP).

Course will convene based on the needs of AOPV command teams. A maximum of two AOPV command teams can challenge the course concurrently due to time and logistical constraints.
All personnel will have taken the basic arctic navigation (BAN) course or PLAR.

Do not need Arctic Operations course to attend this course.

Discussion Point 3:

The board spent a significant amount of time talking about the bridge organization while navigating in ice. The bridge design of an AOPV (Nav 1 bridge) was considered; these considerations were incorporated insofar as possible without any board members having seen the final layout of the bridge. AOPV crews will, in time, identify practices that work, and these best practices will be incorporated into training via this course.

Navigation Organization – including helmsman, possible change to organizational practices. AOPV has Nav1 bridge, which means the bridge can be run by 1 person. Similar to the last discussion point, current RCN configurations may not work well when sailing in ice, and operating procedures will be identified by AOPV 1.

Discussion Point 4:

Train/No Train

T001: Ice covered water needs clarification. There is new equipment that effects this, taught on BAN but a higher proficiency level is required

T002: Taught on basic for discussion.

T009: Report Ice observations: this task will be one the RCN will perform. Who will do it is up for debate, MET Techs are a possibility. They do not have knowledge of navigation and are therefore not the best person to go up in Helos and map the ice. The Navigator may be a better choice. The NavO can chart the passage the Met Tech can draft and submit ice observations to METOC or JOC.

T026: Operate with a CCG icebreaker. Already trained on BAN however the board is covering new topics here in.

Discussion regarding instruction of AOPV stability and engineering:

Not necessarily navigation nor ice related, but not covered on any other courses. Subject material may be covered in CFCDs, command team training and NOPQ. Task put in parking lot, come back, possibly in the introduction to icebreaker. Decided that stability would not be taught however characteristics of an AOPV will be taught. Stability and parametric role will not be explicitly taught on any course (to the knowledge of board members) but does not fit on this course given the focus on navigating in northern latitudes. Perhaps FNO would be a good place to expose at least the NavO to parametric role aboard AOPVs.

Since this course is for the CO, XO, NavO add technical aspects of ship characteristics including damage control (DC) considerations.

T 40-41
Train as part of T26

T42
Possibility of shore support? Touched on in FNO course.

Emergencies, possibility of making a teaching point.
Making sure that paperwork is done completely so that bean counters are happy.
Can we observe for the course?
Possibly in Arctic Ops Course

Discussion Point 5:

The board spent some time discussing the future of both the basic and advanced arctic navigation courses, and are of the opinion that the basic arctic navigation course will eventually fall into NWO 3 and 4, with the advanced course becoming part of the command team training.

Discussion Point 6:

The board used a number of references to build the framework of this QSP. Material in the "Command of an AOPV" was used as a benchmark for knowledge required. The board looked at the QSP from the Deep Draught Navigating Officers course because the target audience and timelines were similar. The board used the "Model Course on Navigation in Polar Waters" as a confirmation tool that necessary topics were covered. Using these references the board decided within a five day program that a single PO would be employed to capture the course content. Leaving the course with the ability to navigate and AOPV through ice covered waters.

PO creation

The board discussed planning a 'navigation' passage vs an 'open ocean' passage, and identified the different planning requirements of each. Given the vast body of water in Canada's Arctic, an open ocean passage was determined to be the best terminology, given the need to convey the numerous considerations for sailing an AOPV in ice covered waters in northern latitudes. Navigation passage seemed too restrictive. The open ocean passage is used as an EO initially, which is designed to include all the considerations, and is used as a basis of the PC to assess the candidates.

Discussion Point 7:

The board deliberated whether RCN vessels will adhere to NORDREG routing message requirements. Initially, it was agreed that RCN vessels were under no obligation to do so. If RCN vessels chose to engage, perhaps PIM messages to RJOC would be most appropriate course of action. As the board progressed, it was discussed that the proven routing messages serve a purpose, and it may behoove the RCN to participate. The CCG provides routes and waypoints to vessels as proven routes from year to year. The individual ship would report to RJOC/MCC with a PIM message who them would provide a message to NORDREG if requested by higher authority. For this reason the creation of the route message was included. Simply understanding requirements of other vessels operating in the North, and elements of a NORDREG routing message may be of value. The CCG proven routing was also included for these reasons.

Discussion Point 8:

The board recognized that most of the required theoretical knowledge regarding ice is taught on the Basic Arctic Navigation course. This course is an opportunity for the CO, XO & NavO to collaborate, develop a plan and execute a voyage through the North in ice. The course was structured to reflect the reality of taking a ship to sea in ice covered waters. Starting with the planning phase, moving into executing the plan, looking into new concepts in ice, such as manoeuvring in ice, freeing a beset vessel and escort operations. The final outcome was a scenario based PO that included parts of theoretical and academic, along with practical ship handling considerations.

Discussion Point 9:

The board spent some time discussing how ships will truly operate in the vicinity of ice. The discussions included the use of helicopters in ice recce, the use of a handheld GPS to plot dangers as well as path of least resistance. The use of radar sat imagery, risk assessment matrix and BRM discussions to support a safe transit through ice were all included. Ice recce in helicopter operations was discussed at length. As discussed previously, the role of an ice observer is a new and unique capability aboard AOPVs. The board identified that taking handheld GPS fixes in the event that helicopters do not have GPS plotters is a fairly basic but essential skill that should not be overlooked. The ice observations should be conducted by someone who understands ice characteristics, ship handling in ice, and ship routing. Those that should fill this position should generally be the CO, XO, or NavO.
Met techs do not forecast ice, and therefore do not need to be trained as ice observers. The board chair emphasized that ice cannot be forecasted accurately.

Discussion Point 10:

The board discussed methods of instruction regarding how to prevent a vessel from being beset in ice. It was identified that due to the possible limitations in the NABS for an AOPV, some topics, such as use of the heeling tanks, would be discussed in a class room and others such as backing and ramming and the star manoeuvre would be performance based. In addition to ship manoeuvring in the simulator the engineering configurations and DC considerations would be verbally confirmed between operator and instructor in the scenarios as required. This course was built understanding the NABS capabilities are not fully known.

Discussion Point 11:

The board chair introduced a risk assessment matrix from Cdr Sea Training Group. Initially it was hoped that this matrix would be applicable to the Risk Assessment EO. Upon further discussion, it became apparent that the risk assessment prior to entering ice needed to be the Cdr Sea Training Group matrix plus. Plus meaning that COs/XOs would naturally perform a risk assessment given elements they are familiar with. Ice presents further risk and COs should be exposed to a risk assessment matrix and process that will allow them to articulate their considerations prior to entering ice. Some discussion points were the bollard test, engineering configuration, fatigue, experience.

Discussion Point 12:

The board discussed method of instruction and whom should instruct the program. It was determined that a FNO qualified instructor would facilitate the course with an AOPV command qualified mentor to lead in the discussions and NABS scenarios. The coast guard provides unique services in Canada that are outside the RCN mandate. It was therefore determined by the board that a guest speaker from the CCG would be required to highlight these unique tasks and the CCG C2 structure.

Initially there was consideration that the CCG representative would accompany students into the simulator upon completion of interactive lectures. However,the RCN currently has a mentorship program for future commanding officers of AOPVs and a MOU between the CCG and RCN to support operations in the Arctic with 4/10ths or greater ice. In addition to this practical experience the RCN has sourced academic training through Memorial University, for commanding officers and executive officers to support introduction of this new capability to the RCN.

This advanced arctic navigation course is intended to replace the academic training received at Memorial University, and the practical experience at sea will evolve to a reciprocal training agreement between the RCN and the CCG once the first AOPV comes into service.

Given the facts discussed here, the CCG will be asked to provide a half day of lectures to discuss the unique tasks executed by the CCG as per the EO 001.03 standard.

Discussion Point 13:

Initial cadre training (ICT) was discussed insofar as whether it would be necessary to teach students about new equipment aboard AOPVs (FLES, Sigma 6, WECDIS). It was ultimately decided that this would all be covered by ICT, thus not necessary. ICT will be provided by Fleetway and Bluedrop. This training consists of equipment training and ship board knowledge that is unique to an AOPV. This

training constitutes the delta training between a CPF and an AOPV. The training was considered for specific AOPV equipment with consideration for the BOQ onboard a CPF.

Discussion Point 14:

The board decided to incorporate some of the functions of the CCG into the course, based on the potential that the AOPV would be obliged to provide assistance to maintain SOLAS, as captured in EO 001.02.

Discussion Point 15:

The board determined that the amount of information that needs to be consumed was so large that a pre-course reading package would be required. The pre-reading package should be broken into references that must be read prior to attending the course, and supplementary references.

Discussion Point 16:

Once the Advanced arctic navigation course reaches a steady state, the RCN will cease sending personnel to memorial university, and the pre-requisite to attend this program will be the basic arctic navigation course as well as a pre-determined practical experience in 4/10ths or greater ice (IMO Polar Code). A PLAR is acceptable when there is sufficient practical experience in ice navigation.

Given the course material the arctic operations course was determined not to be a pre-requisite to the advanced arctic navigation course. However the basic arctic navigation course remains a pre-requisite. The board referenced the arctic operations course and identified potential gaps in requisite knowledge. The minor gaps identified were incorporated into the advanced arctic navigation course.

Discussion Point 18:

The board discussed operating communications and navigation equipment in high latitudes. The board chair had experience operating such equipment in the high latitudes and observed very little difference. That being said, knowledge of the limitations associated with communications equipment in high latitudes needs to be addressed so that issues previously identified in the past are not repeated.

Operations in northern latitudes often require vessels or units to operate beyond mutual supporting distance of each other. Radio transmission and reception devices, while capable of spanning the extended distances, are subject to interference by magnetic storms, aurora borealis, and ionospheric disturbances, which may completely black out reception for hours, or even days. Transmission and reception under most conditions is possible providing proper installation and operation procedures are followed.

Radio communications in the northern latitudes suffer from propagation difficulties and exceptional planning steps are essential for radio operations in northern areas. Emphasis must be placed on the selection and use of proper frequencies and if the radio propagation graphs, charts, and prediction data presently available are fully utilized. In order to appreciate the effect that atmospheric disturbances such as the aurora borealis can have on radio communication, it is necessary to understand the methods by which radio energy from a transmitter can reach the distant radio receiver. The combined effects of distance, terrain, cold, ice and dampness on communications equipment increase maintenance and supply problems to the extent that the full effort of operators and repair personnel is required to provide

satisfactory communications. These knowledge components could be conducted in the Arctic Operations Course. This knowledge component could be covered by NRS and should be covered by the IWO and SNC.

Discussion Point 19:

The Captain's ships books have not yet been created. Upon delivery of HDW the material will be delivered to ADM (mat) and forwarded through the LCMM to the Commanding Officer HDW. Course content will necessarily be further refined once the Captains Ships books are received.

Discussion Point 20:

Competencies were mapped to specific EOs based on prior training and supporting documentation. All supporting competencies were included in each EO to assist with training development.

Discussion Point 21:

For the EC & PC the TDC needs to find/create ice products that show intended ice formations in Admiralty Inlet, or as required by the scenario.

Discussion Point 22:

The board decided to change K009, by removing double ramming procedures. This was done after consultation with a very experienced ice pilot who indicated that this procedure is simply two ice breakers working in tandem clearing ice jams IVO a bridge. A function that the RCN will not likely do.

Discussion Point 23:

The board decided to remove K026 from knowledges after consultation with D Nav (Log) and determined that this was a function of the AOPV Logistics Officer.

Discussion Point 24:

The board determined that chart projections are relevant to arctic navigation but should be captured on basic arctic navigation course.

Lessons Learned:

1. There are significant disparities between the QSP formats used on the west coast and east coast. There was initially confusion as to which QSP format should be implemented for use.

2. When developing QSPs board members would benefit from one shared web based QSP format that would facilitate the development of the QSP.

Guidance to Development and/or Instructional Staff:

1. The course is designed to move from an academic class room setting into scenario driven training. A significant training limitation is the potential inability to access real world ice information (CSI) due to lack of ice. Depending on the time of year, there may not be any real-world ice products available for students to implement into their planning and simulation scenarios. If no appropriate ice products exist the training developers are to create ice products that will support the training. There is limited time in the 5 days to allow for retraining. It is very important during the development of the course to ensure that the time is effectively utilized.

2. When soliciting guest lecturers from the CCG, the preference should always be a current serving sea going captains or chief officer of an icebreaker.

3. PC scenarios should reflect ice and environmental conditions that would be encountered during the shoulders of the navigable season.

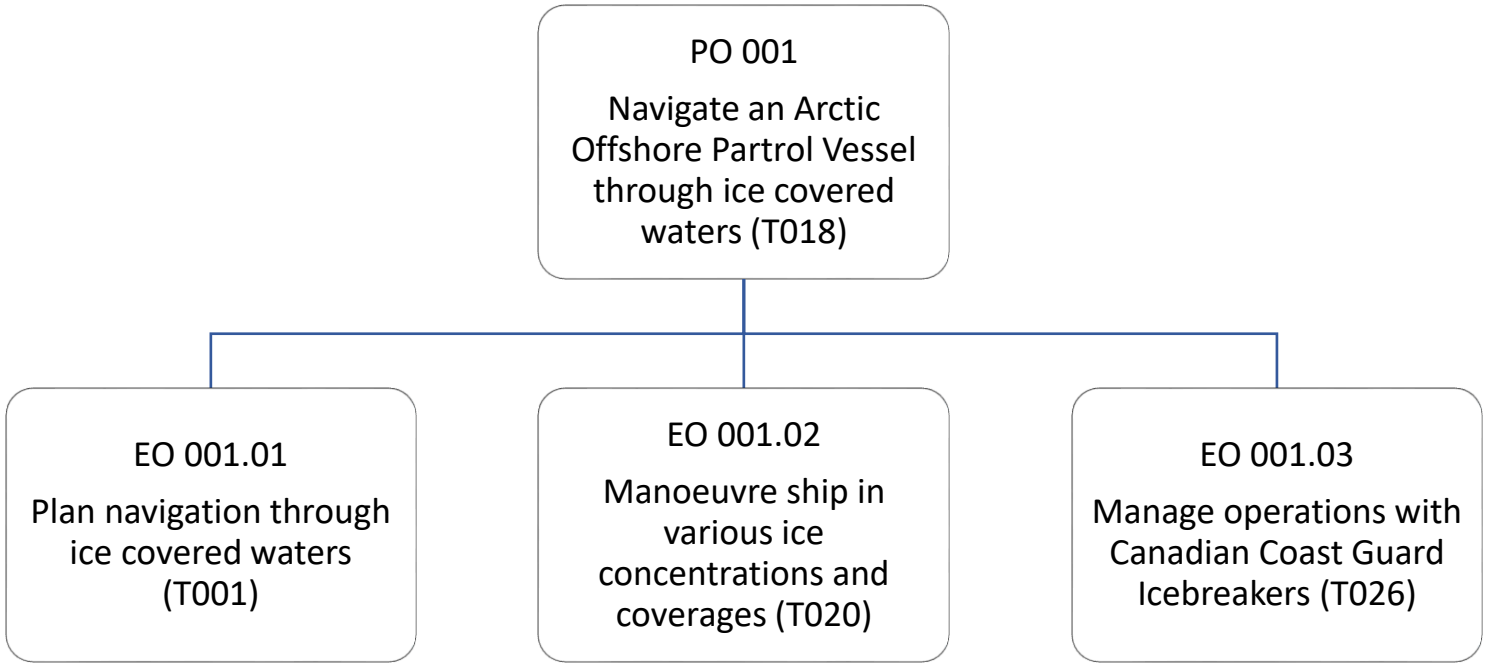
Follow Up Action:

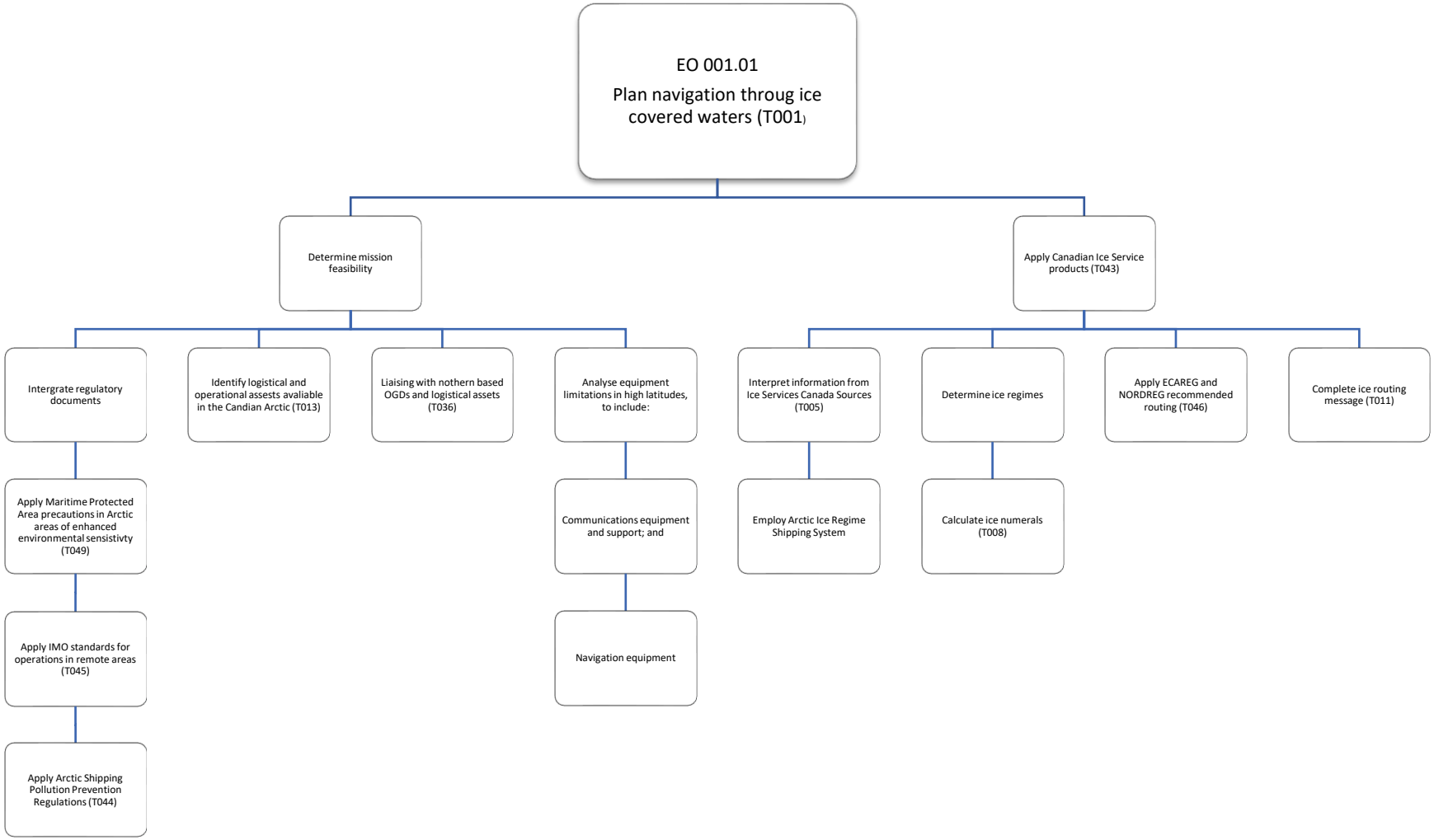
Item	Action	OPI	Expected Date of Completion
1	**TDO Review**	**Lt Powell**	**19 Feb 18**
2	**Add References and Reference Annex, NTDC(P) Review**	**Lt(N) Ronaldson & Lt(N) Zeeman**	**26 Feb 18**
3	**Submit for input XO HDW**	**LCdr Meagher**	**TBD**
4	**Submit for input Captain (retired) Norman Thomas**	**Norman Thomas**	**TBD**
5	**Board Final Review**	**Board Members**	**TBD**
7	**Final Review by Board Chair**	**Cdr Gleason**	**TBD**

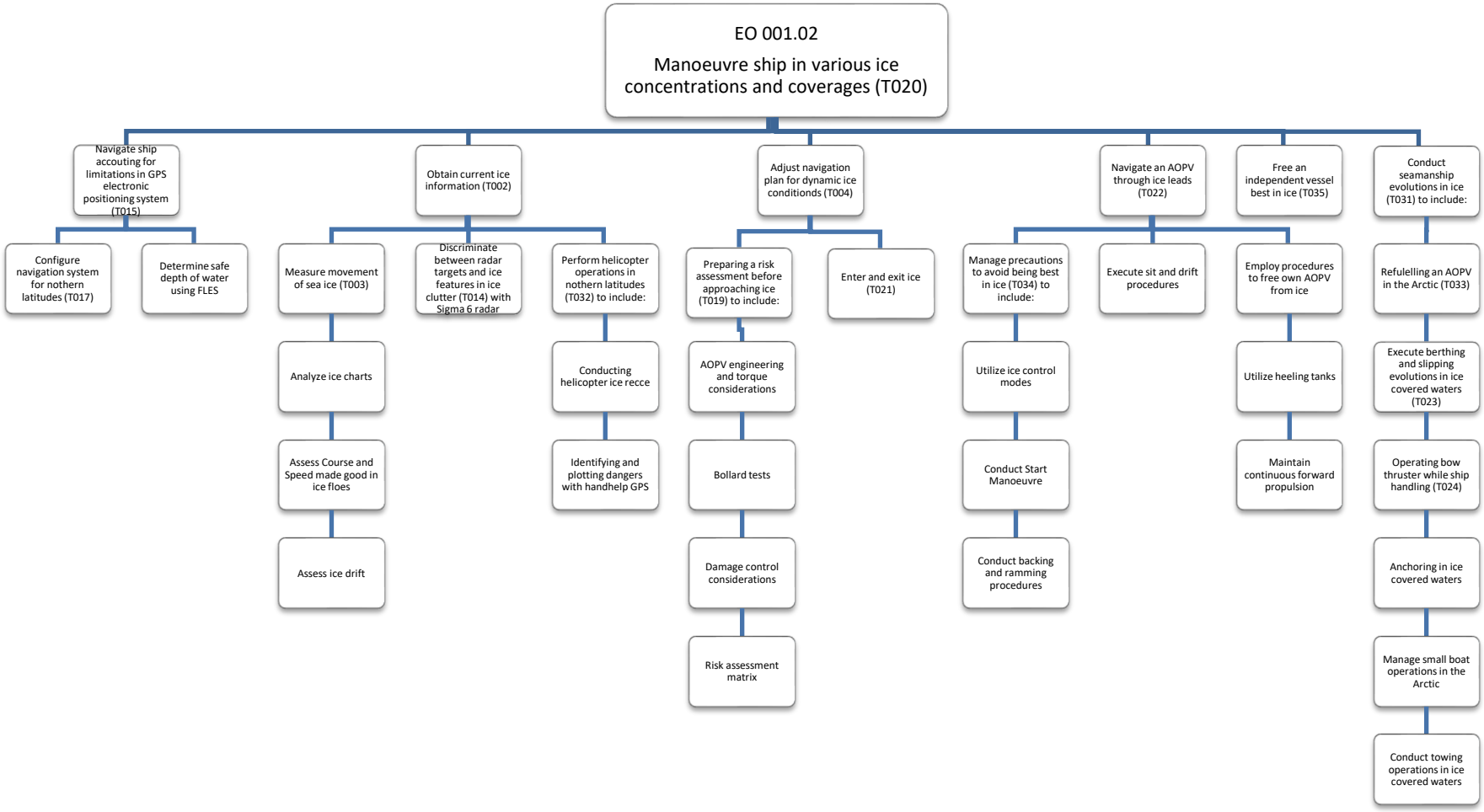
Other Considerations (if applicable):

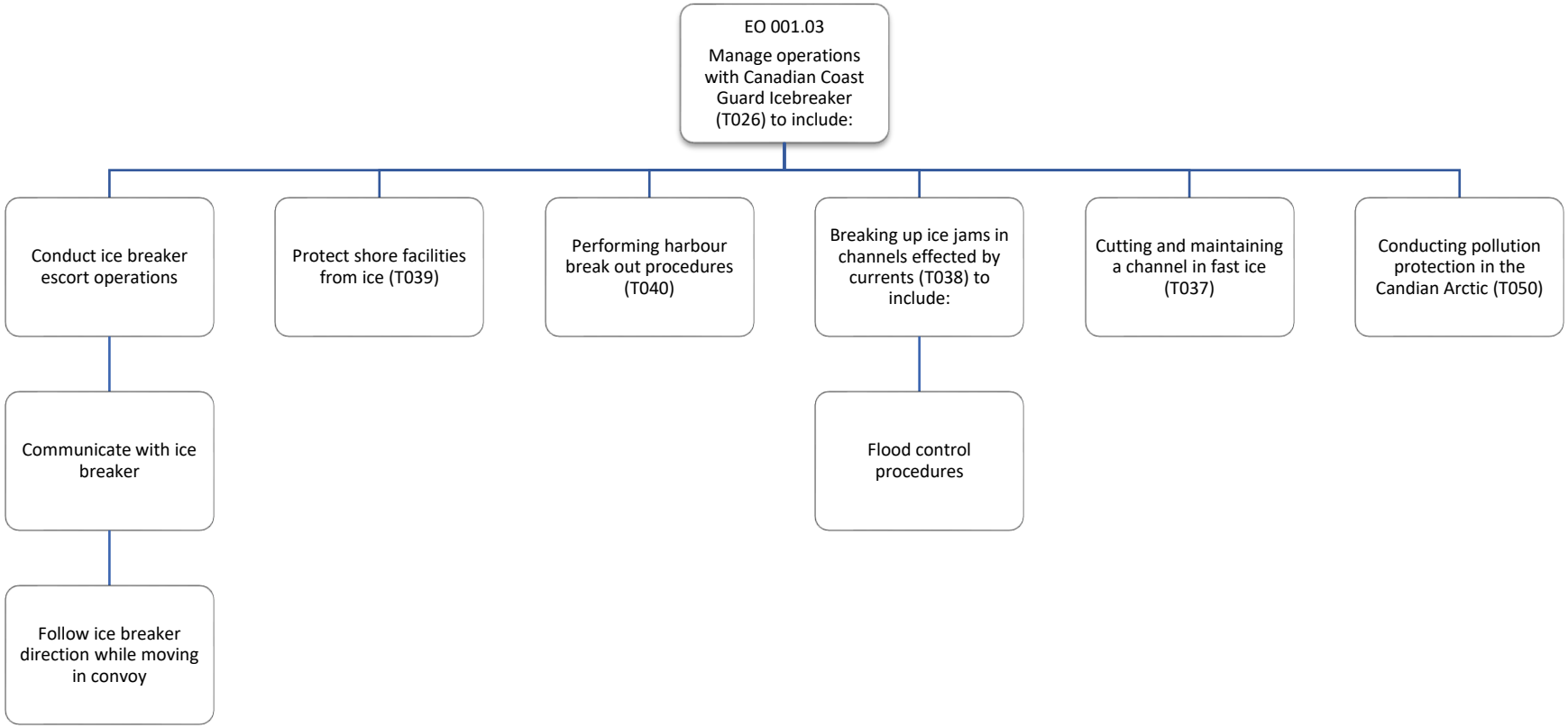
Annex D – Scalar (POs & EOs)

(**Annex D not to be deleted**)




EO 001.01
Plan navigation throug ice covered waters (T001)
Determine mission feasibility
Apply Canadian Ice Service products (T043)
Intergrate regulatory documents
Identify logistical and operational assests avaliable in the Candian Arctic (T013)
Liaising with nothern based OGDs and logistical assets (T036)
Analyse equipment limitations in high latitudes, to include:
Interpret information from Ice Services Canada Sources (T005)
Determine ice regimes
Apply ECAREG and NORDREG recommended routing (T046)
Complete ice routing message (T011)
Apply Maritime Protected Area precautions in Arctic areas of enhanced environmental sensistivty (T049)
Communications equipment and support; and
Employ Arctic Ice Regime Shipping System
Calculate ice numerals (T008)
Apply IMO standards for operations in remote areas (T045)
Navigation equipment
Apply Arctic Shipping Pollution Prevention Regulations (T044)


EO 001.02
Manoeuvre ship in various ice concentrations and coverages (T020)
Navigate ship accouting for limitations in GPS electronic positioning system (T015)
Configure navigation system for nothern latitudes (T017)
Determine safe depth of water using FLES
Obtain current ice information (T002)
Measure movement of sea ice (T003)
Analyze ice charts
Assess Course and Speed made good in ice floes
Assess ice drift
Discriminate between radar targets and ice features in ice clutter (T014) with Sigma 6 radar
Perform helicopter operations in nothern latitudes (T032) to include:
Conducting helicopter ice recce
Identifying and plotting dangers with handhelp GPS
Adjust navigation plan for dynamic ice conditionds (T004)
Preparing a risk assessment before approaching ice (T019) to include:
AOPV engineering and torque considerations
Bollard tests
Damage control considerations
Risk assessment matrix
Enter and exit ice (T021)
Navigate an AOPV through ice leads (T022)
Manage precautions to avoid being best in ice (T034) to include:
Utilize ice control modes
Conduct Start Manoeuvre
Conduct backing and ramming procedures
Execute sit and drift procedures
Free an independent vessel best in ice (T035)
Employ procedures to free own AOPV from ice
Utilize heeling tanks
Maintain continuous forward propulsion
Conduct seamanship evolutions in ice (T031) to include:
Refulelling an AOPV in the Arctic (T033)
Execute berthing and slipping evolutions in ice covered waters (T023)
Operating bow thruster while ship handling (T024)
Anchoring in ice covered waters
Manage small boat operations in the Arctic
Conduct towing operations in ice covered waters


EO 001.03
Manage operations with Canadian Coast Guard Icebreaker (T026) to include:
Conduct ice breaker escort operations
Protect shore facilities from ice (T039)
Performing harbour break out procedures (T040)
Breaking up ice jams in channels effected by currents (T038) to include:
Cutting and maintaining a channel in fast ice (T037)
Conducting pollution protection in the Candian Arctic (T050)
Communicate with ice breaker
Flood control procedures
Follow ice breaker direction while moving in convoy

Annex E – OJPR

(**Annex E not to be deleted**)

Not applicable for this iteration of the QSP for FNO
See Annex N for the Pre-Instruction Package (PIP)

Annex F – Acronyms, Abbreviations & Terms

(**Annex F not to be deleted**)

This annex is composed of two (2) parts:

- Part 1 will detail acronyms and abbreviations used in this QSP.
- Part 2 will detail terms used in this QSP.

Part 1 - Acronyms and Abbreviations

Acronym or Abbreviation	Term or Definition
AAN	Advanced Arctic Navigation Course
AIRSS	Artic Ice Regime Shipping System
AOPV	Arctic Offshore Patrol Vessel
AOR	Area of Responsibility
ASPPR	Arctic Shipping Pollution Prevention Regulations
ATT	Actual Training Time
BAT	Base Administrative Time
BAN	Basic Arctic Navigation Course
BRM	Bridge Resource Management
CAC	Canadian Arctic Category
CAF	Canadian Armed Forces
CIS	Canadian Ice Services
CJOC	Canadian Joint Operations Command
CFCD	Canadian Forces Controlled Document
CCG	Canadian Coast Guard
CHS	Canadian Hydrographic Service
COY	Company
CPF	Canadian Patrol Frigate
C/NPTG	Commander Naval Personnel and Training Group
CFITES	Canadian Forces Individual Training and Education System
CO	Commanding Officer
CO NTDC (A)	Commanding Officer Naval Training Development Center Atlantic
CONOPS	Concept of Operations
CONUSE	Concept of Use
CSTG	Commander Sea Training Group
DC	Damage Control
DCC	Damage Control Condition
DFO	Department of Fisheries and Oceans
DG	Diesel Generator
DLN	Defence Learning Network
EC	Enabling Check
ECPINS	Electronic Charting and Position Indicating System
EDI	Explain Demonstrate Imitate
ECAREG	Eastern Canada Vessel Traffic Service System
FNO	Fleet Navigating Officer

Acronym or Abbreviation	Term or Definition
FLES	Forward Looking Echo Sounder
EO	Enabling Objective
GPS	Global Positioning System
GMDSS	Global Maritime Distress and Safety System
HDW	Harry DeWolf
IACS	International Association of Classification Societies
IAW	In Accordance With
ICT	Initial Cadre Training
IMO	International Maritime Organization
JOC	Joint Operations Command
JTFN	Joint Task Force North
LEDET	Law Enforcement Detachment
MARPIF	Maritime Port Information
MCTS	Maritime Communications and Traffic Services
METOC	Meteorological and Oceanographic Centre
MITE	Military Individual Training and Education
MLA	Mean Line of Advance
MODIS	Moderate Resolution Imaging Spectroradiometer
MOU	Memoranda of Understanding
MPA	Maritime Protected Area
MCC	Maritime Component Commander
MRRB	Multi-Role Rescue Boat
NABS	Navigation and Bridge Simulator
NFP	Notice for Power
NFS (A)	Naval Fleet School Atlantic
NFS (P)	Naval Fleet School Pacific
NOPQ	Naval Officer Professional Qualification
NNF	Nanisivik Naval Facility
NFR	Naval Force Readiness
NORDREG	Northern Canada Vessel Traffic Service System
NPTG	Naval Personnel and Training Group
NQual	National Qualification
NTDC (A)	Naval Training Development Centre Atlantic
NTDC (P)	Naval Training Development Centre Pacific
NWO	Naval Warfare Officer
NWP	North West Passage
OFP	Operational Function Point
OGD	Other Governmental Department
OOW	Officer of the Watch
PC	Performance Check
PC	Polar Class
PG&D	Power Generation and Distribution
PI	Parallel Index Line
PIM	Plan of Intended Movement
PLAR	Prior Learning Assessment and Recognition

Acronym or Abbreviation	Term or Definition
PO	Performance Objective
PPE	Personal Protective Equipment
PRB	Progress Review Board
PT	Physical Training
QSP	Qualification Standard and Plan
RACON	Radar Beacon
RCN	Royal Canadian Navy
RJOC	Regional Joint Operations Command
SOM	Scheme of Manoeuvre
SOP	Standard Operating Procedure
SEMS	Safety and Environmental Management System
SOLAS	Safety of Life at Sea
RQ	Rank Qualification
TA	Training Authority
TAT	Training Administrative Time
TE	Training Establishment
TP	Training Plan
TRB	Training Review Board
TTT	Travel training Time
UAS	Unmanned Aircraft System
USCG	United States Coast Guard
WECDIS	World Electronic Charting and Display System
XO	Executive Officer

Part 2 – Terms

Not used in this QSP

Term	Definition

Annex G – Suggested Sequencing & Timings

(**Annex G not to be deleted**)

Part 1 – Scheduling Resources

Activity or Assessment	Enabling or Performance Objective	Personnel Resources	Required Resources (for scheduling)	Trg Time/resource (mins)	EC Time (mins)	CEC/PC Time (mins)
PO 001: Navigate a vessel					Total PO time:	1390
PO ATT:		PO Admin Time:		PO PT time:	0	
EC 001/002					210	
CEC					300	

Actual Training Time	ATT	1900
Base Administrative Time	BAT	50
Training Administrative Time	TAT	50
Training Travel Time	TTT	0
Physical Fitness Training	PT	0
	Total	2000
	Total Days	5
	Rounded	5

Part 2 - Ideal Course Schedule

The ideal course schedule should be read in conjunction with the Activity and Assessment Plan at Annex K.

07:30	0800 – 0900	0900 – 1000	1000 – 1100	1100-1200	1300 – 1400	1400 – 1500	1500 – 1600	Self - study
Day 1	**Introduction** Admin time Course purpose Pre-reading documents Intro to the Polar Code Cyclical planning cycle	**AOPV characteristics** Ship design and capability Polar Code classification system	**Systems in the North** ECAREG / NORDREG ASPPR vs AIRSS MCTS in the north Ice Regime	**Ice services products** Finding products CCG proven routing Tidal ranges in North MODIS software Ice routing message	**Transiting the North** NW passage – 7 routes Iceberg Alley Seasonal variability	**Arctic Ops** Hamlets Med supply Resupply Fueling Embarkation / Disembarkation MPAs Arctic Ops Cultural/Aboriginal awareness training ID northern based Logistic assets		Intro to the EC
Day 2	**Nav equipment in high latitudes** FLES Sigma 6 GPS vs surveys Use of PI's RACON Radar tuning	**Identifying a path through ice** Path of least resistance Use of a Helo Ice Vue	**Using Ice services products** Exercise to determine sea ice extent and Ice Regime Reporting current ice	**Enter/Exit Ice** Introducing hull into Ice Speed vs Power Appropriate route Intro to risk assessment	**NABS - Configure Nav systems** FLES Datums Ice echoes for different ice Use of RACON	**EC 002 – Phase 1 Driving in Ice (NABS)** Enter / Exit ice Backing and ramming Star manoeuvre		
Day 3	**Manouevring in ice** Sit & drift procedures Free own AOPV beset in ice	**Rendering assistance** Evaluating tasking Rendering assistance	**EC 002 – Phase 2 Rendering assistance in Ice (NABS)** Tasking to render assistance Breakout procedure Taking up the escort		**Shiphandling** Fueling: NNF/CCG vessel/Barge Berthing and slipping Bowthrusters	**EC 002 – Phase 3 Alongsides/Departures in ice (NABS)** Berthing/Slipping Coming alongside another ship at anchor		Case Study – CCG towing
Day 4	**Shiphandling & Boats (Case Study)** Towing Anchorage Small boat (MRRP) Anchorage		EC Brief prep time Flex time	EC 001 – Open ocean passage brief – Iqaluit - NNF	EC 003 – Guest Lecture Canadian Coast Guard CO - Protect shore facilities from ice - Break up ice jams in channels affected by current - Conduct pollution protection in Cdn Arctic - Ice breaker ops			
Day 5	**Update Brief / Plan** Based on new information & location	**PC 001 – Navigate an AOPV thru Ice covered waters**			PC 001 Team 2 Retrain / Retest / Admin			

Annex H – Additional Training Resources

(**Annex H not to be deleted**)

This annex is a compliment to the staffing and critical resources listed in chapter 1 of this QSP

FC = Full Course, A/R = As Required

STORES REQUIREMENTS		
Item Description	Quantity	Notes

Annex I – Costing Information

(Annex I not to be deleted)

Estimates only, based on CF Cost Factors Manual FY 2012/13

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: TDL 300		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	14	$1,000.0		$14,000	$14,000	$70,000.00
R&Q	14	$36.0	20	$10,080	$10,080	$50,400.00
Incidentals	14	$17.5	20	$4,900	$4,900	$24,500.00
Subtotal				**$28,980**	**$28,980**	**$144,900.00**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	16	$50.0	1	$800	$800	$4,000.00
Classroom (square meters)	30	$13.00	15	…	…	…
Computer requirements	16	$1,000.0	1	$16,000	$0	$16,000.00
IT Equip (Proj, Smart board etc)	1	$6,000.0	1	$6,000	$0	$6,000.00
DWAN connection (Set up)	1	$5,000.0	1	$5,000	$0	$5,000.00
Link Toolkit (on DVD)	15	$1.0	1	$15	$15	$75
Instructional Materials (Printed / Handouts etc)	15	$30.0	1	$450	$450	$2,250.00
Subtotal				**$28,265**	**$1,265**	**$33,325.00**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Instructor (PO1/WO) (incremental)2	2	$223.0	20	$8,920	$8,920	$44,600.00
Incremental Civilian Instructor		XXX				
		subtotal		**$8,920**	**$8,920**	**$44,600.00**
	Total Estimated Costs			**$66,165**	**$39,165**	**$222,825**

Annex J – Course History

(**Annex J not to be deleted**)

It must be annotated with every update or modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Details on course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications shall be captured in the table below. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Link to QSP amendment form (Annex E of NPTORD 4500-22)

NPTORD 4500-22
QSP - Amendment -

Reason for amendment (with date)	Details on course history
	AAN RDIMS 1021714v2

Annex K – Assessment Plan

(Annex K not to be deleted)

Part 1 - Assessment Overview

This assessment plan shall be read in conjunction with the Annex G Part 2 Ideal Course Schedule

EC/PC	Scope Covered	Purpose	How	Pass/Fail Criteria
EC 001.01	EO 001.01	To assess student's ability to plan a navigation through ice covered waters	Through practical assessment. Students will present their navigation plan in a formal briefing	Must achieve a SAT in all critical items and not more than 3 UNSAT in non-critical items
EC 001.02	EO 001.02	To assess a student's ability to manoeuvre a ship through various ice concentrations and coverages	Through practical assessment of student in simulator in 3 phased EC	Must achieve a SAT in all critical items and not more than 3 UNSAT in non-critical items
PC 001	EO 001.01, EO 001.02 and EO 001.03	To assess a student's ability to navigate an AOPV through ice covered waters	Through practical assessment of student in simulator	Must achieve a SAT in all critical items and not more than 3 UNSAT in non-critical items

Part 2 - Activity and Assessment Plan (AAP)

AAP was not used for this QSP; however, draft assessment plans are shown below:

Assessment Plan for EC 001.01 - Brief a navigation plan through ice covered waters.
Activity:
The student is required to present a navigation plan (via formal brief) through ice covered waters.
Input:
Instructor gives an operational briefing outlining the AOPV's tasking/mission.
Outcome:
An executable arctic passage plan delivered through briefing (Not a track by track).
Feedback:
Instructor led feedback, with peer to peer feedback if available based on course load.
Scenario:
A typical OP NANOOK tasking based on:
a) Canadian Joint Operations Command (CJOC) Commanders Planning Guidance – OPERATION NANOOK 20xx;
b) Maritime Component Commander (MCC) OP NANOOK 20xx – Maritime exercise initial planning conference administrative instructions; and
c) OP NANOOK Scheme of Manoeuvre (Instructor to make every effort to get the slide for the next years OP NANOOK, failing that use one of four SOM on course reference material).
Resources:
Chart outfit (electronic and paper);
Chart catalogue;
WECDIS capable computer;
Navigation planning tools;
Ice charts and meteorological data;
Reference material;
Internet access;
Classroom and associated equipment; and
Standard FNO open ocean briefing template.

Conditions:

Classroom setting.

Limitations:
Time constraints;
Real time sourcing of ice information; and
No ability to liaise with outside stake holders.

Impact:
Review course material, and thorough debrief of deficient areas.

Assessment Plan for EC 001.02: Manoeuvre Ship in Various Ice Concentrations and Coverages

Activity:

The student is required to manoeuvre a ship in various ice concentrations and coverage.

Phase 1: Navigating in coastal waters, entering exiting Ice, navigating through ice leads

Input:

Scenario, for entering ice in an AOPV.

Outcome:

Student successfully identifies risk, and completes risk assessment, and safely enters/exits ice.

Feedback:

Instructor (throughout with formal debrief on completion), and self-assessment (bridge team).

Phase 2: Freeing a beset vessel/escort

Input:

Scenario with beset vessel.

Outcome:

Student successfully frees beset vessel and safely brings it under escort.

Feedback:

Instructor (throughout with formal debrief on completion), and self-assessment (bridge team).

Phase 3: Conducting seamanship evolutions

Input:

Scenario with seamanship evolutions to include:
Alongside/Departures (NNF, Quebec, Q.C, Halifax N.S);
Alongside CCG vessel at anchor; and
Anchoring (in ice field).

Outcome:

Student safely conducts seamanship evolutions in ice.

Feedback:

Instructor (throughout with formal debrief on completion), and self-assessment (bridge team).

Scenario:

Based on input.

Resources:

Ice products;
Bridge team;
NABS;
Classroom;
Mentor (current/post AOPV command);
Scenarios; and
Support manning.

Conditions:

Ice covered waters;
Day/Night; and
Varied environmental conditions.

Limitations:

Simulator capabilities; and
Database - ship model, graphics, visual ice modeling, modeling of ice interactions (pressure, friction, etc.).

Impact:

Non-critical EC, feedback and retrain.

<u>Assessment Plan for PC 001</u> - Navigate an AOPV through ice covered waters
Activity:
The student is required to navigate an AOPV through ice covered waters.
Input:
Tasking that will force the ship to transit through 10/10 medium first year ice which may include old ice inclusions. The scenario and ship's routing will coincide with EC 001.01.
Outcome:
Student will successfully navigate an AOPV through ice covered waters, to include:
Selecting an appropriate route for AOPV based on ice regime;
Freeing a beset vessel; and

Proceeding alongside in ice covered waters.

Feedback:
Instructor based feedback.
Scenario:
AOPV transiting from Iqaluit to NNF. Transiting through Admiralty Inlet they come upon a beset vessel that has recently departed Arctic Bay bound for Pond Inlet. Beset vessel/RJOC to engage AOPV for assistance (Type A vessel- positive ice regime too much pressure). Upon freeing vessel they proceed to NNF to remove the remaining DFO from the tanks. Alongside position will be initially fouled by brash ice.
Resources:
Ice products;
Bridge team;
NABS (configured for AOPV);
Classroom;
Scenarios;
Support manning;
Instructor; and
Exercise instructions.

Conditions:
Ice covered waters;
Day/Night; and
Varied environmental conditions.

Limitations:
Simulator capabilities;
Database, ship model, graphics, visual ice modeling, modeling of ice interactions (pressure, friction, etc.);
Time constraints; and
Inputs artificially required to progress through the scenario.

Impact:
Remedial training; and/or

Re-test.

Annex L – TEL Plan

(**Annex L not to be deleted**)

Information about development of Technology Enabled Learning (TEL) can be found in the following embedded document:

Spirals.docx

Application	COA 1 TEL Delivery	COA 2 Interim Delivery	Impact if TEL not implemented - Realized benefits - E/W Coast Delta	Priority

Annex M – Target Population Analysis & Training Establishment Details

(**Annex M not to be deleted**)

Was not done for this QSP

<u>Key Target Population Details</u> – (identify the candidates/students/trainees for the course)

"The learners' level of skill and knowledge with respect to the performance objectives (POs) is a factor which depends mainly on previous related training or education, and experience. It may also be affected by strong opinions or attitudes about the subject matter. Information should be obtained on the learners" - CFITES Vol 4

Annex N – Pre-Instruction Package (PIP)
(**Annex N not to be deleted**)

Not applicable to this QSP